ADVANCED MULTIPLE ACCESS TECHNIQUES FOR BACKSCATTER-BASED COMMUNICATION

A PROJECT REPORT

submitted by

Amina N S (MBT23EC024 / B23EC1213)
Ardra A Nair (MBT23EC030 / B23EC1217)
Arya R Nair (MBT23EC034 / B23EC1218)
Daya R S (MBT23EC044 / B23EC1223)

to

the APJ Abdul Kalam Technological University
in partial fulfilment of the requirements for the award of the Degree of

BACHELOR OF TECHNOLOGY
IN
ELECTRONICS AND COMMUNICATION ENGINEERING



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
MAR BASELIOS COLLEGE OF ENGINEERING & TECHNOLOGY (Autonomous)
Mar Ivanios Vidyanagar, Nalanchira
Thiruvananthapuram
November 2026

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
MAR BASELIOS COLLEGE OF ENGINEERING & TECHNOLOGY (Autonomous)
Mar Ivanios Vidyanagar, Nalanchira, Thiruvananthapuram.

CERTIFICATE

This is to certify that the report entitled **"Advanced Multiple Access Techniques for Backscatter-Based Communication"** is a bona fide record of the project work carried out by **Amina N S (MBT23EC024 / B23EC1213), Ardra A Nair (MBT23EC030 / B23EC1217), Arya R Nair (MBT23EC034 / B23EC1218)** and **Daya R S (MBT23EC044 / B23EC1223)** of the seventh semester, Electronics and Communication Engineering Branch, towards the partial fulfilment of the requirements for the award of the **Bachelor of Technology Degree in Electronics and Communication Engineering** of the **APJ Abdul Kalam Technological University, Kerala**. This work was carried out by them under our guidance and supervision. This report, in any form, has not been submitted to any other University or Institute for any purpose.

Dr. Sreedevi P	**Dr. Sreedevi P**	**Dr. Luxy Mathews**
(Project Co-ordinator)	(Internal Supervisor)	(Head of the Department)
Associate Professor	Associate Professor	Associate Professor
Department of ECE	Department of ECE	Department of ECE

Place: Thiruvananthapuram
Date:

External Examiner

ACKNOWLEDGEMENT

Completing this main project has been a demanding and deeply rewarding experience for us. Along the way we have gained a great deal from the guidance, encouragement and advice of many people, and we sincerely thank each of them.

We express our deepest gratitude to **Dr. Luxy Mathews**, Associate Professor and Head of the Department of Electronics and Communication Engineering, for her valuable advice and constant support, which helped us to improve our work.

We are sincerely thankful to our internal guide and Project Coordinator, **Dr. Sreedevi P**, Associate Professor, Department of Electronics and Communication Engineering, MBCET. Her steady support, timely feedback and careful supervision kept the project on course from the beginning to the end.

We owe special thanks to our project guides at the Indian Institute of Space Science and Technology (IIST), **Dr. Vineeth B. S.**, Associate Professor, Department of Avionics, IIST, and **Dr. Basudev Majumdar**, Associate Professor, Department of Avionics, IIST. Their expert guidance, technical insight and patient mentoring were central to the successful completion of this project, and we are grateful for the opportunity to carry out our work at IIST.

We are grateful for the blessings and grace of the Almighty, which gave us the strength and determination to complete this work.

Finally, we thank our parents, friends and classmates for always standing by us and for their helpful advice and support throughout this project. We are thankful to everyone who helped us, directly or indirectly, in completing this project successfully.

ABSTRACT

Backscatter communication is a low-power wireless technique in which a device conveys information by modulating and reflecting an incident radio-frequency (RF) signal instead of generating its own carrier. As a result, it uses less energy and simpler, cheaper hardware, which suits Internet of Things (IoT) applications and battery-free devices. When many tags share one reader, however, reliable operation is limited by interference, packet collisions and the difficulty of separating the transmissions of individual tags.

This project addresses these issues through four complementary workstreams. A switch-loaded tag was designed in KiCad around an ADG902 RF switch, with a four-pin connector through which an external ESP32-S3 controller supplies power and the switching control signal. The tag design was imported into CST Studio Suite to study its backscattering behaviour, and the simulated radar cross-section (RCS) at 220 MHz is approximately $+1.5$ dBsm in the ON state and -16.7 dBsm in the OFF state, a contrast of about 18.2 dB between the two load conditions.

For signal-level analysis, a three-tag backscatter model was built in GNU Radio Companion. Each tag modulates its data onto a distinct subcarrier, the tag responses are combined with Gaussian noise, and each receiver branch recovers its tag's bits through band-pass filtering, envelope detection, matched filtering, symbol synchronisation and slicing. In the baseline configuration we obtained bit error rates (BER) of 0.000025, 0.000013 and 0.000019 for Tags 1, 2 and 3. A noise sweep then gave the BER-versus-SNR behaviour of all three tags, with the BER rising steadily as the noise power was increased.

At the packet level, Irregular Repetition Slotted ALOHA (IRSA) with Successive Interference Cancellation (SIC) was implemented independently in Python and evaluated by Monte Carlo simulation over a frame of 200 slots. With the repetition-degree distribution $\Lambda(x) = 0.5x^2 + 0.28x^3 + 0.22x^8$, the scheme reached a peak throughput of 0.7714 decoded packets per slot at an offered load of 0.80, against 0.3678 for Frame Slotted ALOHA simulated under identical conditions. This packet-level study was carried out separately from the GNU Radio model and the tag hardware. Integrating the four workstreams, fabricating the PCB and validating the complete link over the air are the next stages of the work.

CONTENTS

1 Introduction **1**

1.1 Background 1

1.1.1 Low-Power Wireless Devices and the Cost of Active Transmission . . 1

1.1.2 Principle of Backscatter Communication 1

1.1.3 Multiple Access in Communication Systems 3

1.1.4 Multi-Tag Backscatter and the Base Work 4

1.2 Problem Statement 4

1.3 Motivation 5

1.4 Aim of the Project 5

1.5 Objectives of the Project 5

1.6 Scope of the Project 6

1.7 Overview of the Methodology 7

1.8 Organisation of the Report 7

2 Literature Review **8**

2.1 Evolution of Low-Power Wireless Communication 8

2.2 Fundamentals of Backscatter Communication 8

2.3 Review of the Base Work: Ambient Backscatterers for Low-Cost and Low-Power Wireless Applications 9

2.3.1 Backscatter Principle and Received-Signal Model 9

2.3.2 Subcarrier Generation and Frequency-Division Access 10

2.3.3 Tag Architecture 11

2.3.4 Receiver Processing Chain 11

2.3.5 Ambient FM Backscatter and Error Analysis 12

2.3.6 High-Order Modulation and Future Directions 12

2.3.7 Relation of the Base Work to the Present Project 12

2.4 Load Modulation and Reflection Coefficient 13

2.5 Antenna Behaviour and Radar Cross-Section 14

2.6 Backscatter Link Budget 14

2.7 Receiver Processing in Backscatter Systems 15

2.8 Challenges in Multi-Tag Communication 15

2.9 Multiple-Access Techniques 16

2.10 Slotted ALOHA 16

2.11 Frame Slotted ALOHA 17

2.12 Contention Resolution Diversity Slotted ALOHA 17

2.13 Irregular Repetition Slotted ALOHA . . . 18
2.14 Successive Interference Cancellation . . . 19
2.15 Performance Evaluation Metrics . . . 19
2.16 Research Gap . . . 19

3 System Architecture . . . 21
3.1 System Overview . . . 21
3.2 System Architecture Diagram . . . 22
3.3 Functional Requirements . . . 23
3.4 Non-Functional Requirements . . . 24
3.5 Hardware Architecture and Requirements . . . 24
3.6 Software Architecture and Requirements . . . 24
3.7 Technical Specifications . . . 25
3.7.1 Key Parameters . . . 25
3.7.2 Differences from the Base Work . . . 25
3.8 Design Constraints and Assumptions . . . 26
3.9 Feasibility Analysis . . . 26
3.10 System Boundary . . . 27

4 System Design and Methodology . . . 28
4.1 Design Overview . . . 28
4.2 Hardware Design . . . 28
4.3 Electromagnetic Analysis Design . . . 28
4.4 Signal-Level Modelling Design . . . 30
4.5 Packet-Level Multiple-Access Design . . . 30
4.6 Software Tools . . . 31
4.7 Simulation Parameters and Assumptions . . . 31
4.7.1 CST Electromagnetic Simulation . . . 31
4.7.2 GNU Radio Signal Model . . . 32
4.7.3 Standalone IRSA/SIC Simulation . . . 33
4.8 Separation of the Simulation Methodologies . . . 34
4.9 Performance Metrics . . . 34

5 Implementation . . . 36
5.1 Implementation Environment and Setup . . . 36
5.2 KiCad Hardware Implementation . . . 36
5.2.1 Power and Control Pathways . . . 36
5.2.2 PCB Layout and 3D Model . . . 37
5.3 CST Electromagnetic Simulation . . . 37
5.4 GNU Radio Signal Model . . . 38

5.4.1 Baseband Data Generation . . . 39
5.4.2 Signal Multiplication, Scaling and Combining . . . 39
5.4.3 Power and SNR Measurement . . . 41
5.4.4 Receiver Processing . . . 42
5.4.5 BER Evaluation . . . 44
5.5 Standalone Implementation of IRSA and SIC . . . 45
5.5.1 Software Structure . . . 46
5.5.2 IRSA Packet Repetition and Slot Allocation . . . 47
5.5.3 Collision Modelling and SIC Decoding . . . 47
5.5.4 Throughput Calculation . . . 49
5.5.5 Baseline Schemes under Identical Conditions . . . 50
5.6 Implementation Status . . . 50
5.7 Testing and Verification Methodology . . . 50
5.8 Implementation Challenges and Solutions . . . 51

6 Results and Discussion . . . 52
6.1 Experimental Setup and Test Conditions . . . 52
6.2 PCB Design Outcome . . . 52
6.3 CST RCS Results . . . 52
6.4 GNU Radio Signal-Level Results . . . 53
6.4.1 Power and SNR Measurement . . . 53
6.4.2 Received Spectrum . . . 54
6.4.3 Time-Domain Signals and Delay Alignment . . . 55
6.4.4 Recovered Symbol Outputs . . . 55
6.4.5 BER Results . . . 56
6.5 Throughput Evaluation of the Standalone IRSA–SIC Simulation . . . 58
6.6 Comparison with Baseline Methods . . . 59
6.7 Discussion of the Throughput Behaviour . . . 61
6.8 Discussion: Objectives, Advantages and Limitations . . . 62
6.8.1 Achievement of the Objectives . . . 62
6.8.2 Advantages of the Proposed Approach . . . 63
6.8.3 Limitations of the Present Results . . . 63

7 Conclusion and Future Scope . . . 64
7.1 Summary of the Completed Work . . . 64
7.2 Major Findings . . . 65
7.3 Major Contributions of the Project . . . 65
7.4 Overall Conclusion . . . 66
7.5 Constraints in the Current Work . . . 66
7.6 Future Scope . . . 66

7.6.1 PCB Fabrication and Hardware Testing 66
7.6.2 Integration with the Backscatter RF Set-Up 66
7.6.3 Validation and Extension of the GNU Radio Model 67
7.6.4 Integration of IRSA and SIC with the Backscatter System 67
7.6.5 Evaluation of Multiple-Access Performance 67
7.6.6 Physical Multi-Tag Demonstration 67
7.7 Potential Applications 68
7.8 Final Remarks 68

REFERENCES 69

APPENDICES 71
A Summary of Evidence Level 71
B Commands to Reproduce the Results 71
C Source Listings 72

LIST OF FIGURES

Fig 1.1 Conceptual overview of a backscatter communication link 2

Fig 1.2 Comparison between conventional active transmission and backscatter communication 3

Fig 1.3 Overall project approach 7

Fig 2.1 Bistatic backscatter principle with emitter, tag and reader channels. Source: Daskalakis[1]. 9

Fig 2.2 Subcarriers produced by two tags switching at different frequencies around a carrier F_c 10

Fig 2.3 Two-state load switching at a backscatter tag 14

Fig 2.4 Slotted ALOHA access principle 16

Fig 2.5 Frame Slotted ALOHA access principle 17

Fig 2.6 IRSA replica-transmission concept 18

Fig 3.1 System architecture of the project: four parallel workstreams and the proposed integration stage 22

Fig 4.1 KiCad PCB layout and 3D model of the RF backscatter tag 29

Fig 4.2 System-level interface between the computer, the external ESP32-S3 controller and the RF tag PCB through connector J1 29

Fig 5.1 Schematic of the RF backscatter tag with the regulator, level translator, RF switch and connector interface 37

Fig 5.2 CST simulation set-up and model configuration of the dipole tag 38

Fig 5.3 CST-derived RCS versus frequency for the ON-state (2 Ω) and OFF-state (0.45 pF) load conditions 38

Fig 5.4 Overall structure of the three-tag GNU Radio signal model (simplified block diagram) 39

Fig 5.5 Carrier and control-signal generation for the three simulated tags 40

Fig 5.6 Baseband data generation using Vector Source, Char-to-Float and Interpolating FIR filter blocks 40

Fig 5.7 Signal multiplication, tag-response modelling, scaling, noise addition and signal combining 41

Fig 5.8 Combined-signal output and power/SNR measurement branches in the flowgraph 42
Fig 5.9 Tag 1 receiver-processing flowgraph 43
Fig 5.10 Tag 2 receiver-processing and BER-evaluation flowgraph 44
Fig 5.11 Tag 3 receiver-processing and BER-evaluation flowgraph 44
Fig 5.12 Tag 1 BER-evaluation flowgraph with reference generation, delay alignment and the BER block 45
Fig 5.13 Flow of the standalone IRSA/SIC simulation 46
Fig 5.14 Example frame for SIC decoding: shaded cells are packet replicas, c_s is the initial slot occupancy 48

Fig 6.1 Number Sink outputs of the baseline simulation: BER of Tags 1–3, backscatter power, noise power and instantaneous SNR 53
Fig 6.2 Received spectrum of the Tag 1 branch with lines at $\pm$11 kHz and at the $\pm$10 kHz carrier frequency 54
Fig 6.3 Received spectrum of the Tag 2 branch after band-pass filtering, with lines at $\pm$10.8 kHz 54
Fig 6.4 Received spectrum of the Tag 3 branch after band-pass filtering, with lines at $\pm$11.2 kHz 54
Fig 6.5 Time-domain signals of the Tag 1 branch (control frequency 1000 Hz): input data and receiver-stage outputs 55
Fig 6.6 Time-domain signals of the Tag 2 branch (control frequency 800 Hz) 55
Fig 6.7 Time-domain signals of the Tag 3 branch (control frequency 1200 Hz) with the delay-alignment control 56
Fig 6.8 Recovered symbol samples at the output of the Symbol Sync block for Tags 1, 2 and 3 56
Fig 6.9 BER versus SNR for the three-tag backscatter system obtained from the GNU Radio model 57
Fig 6.10 IRSA with SIC: packet loss rate (left) and throughput (right) versus channel load G ($N = 200$ slots, $\Lambda(x) = 0.5x^2 + 0.28x^3 + 0.22x^8$, 800 frames per point) 58
Fig 6.11 Throughput versus offered load for IRSA with SIC, CRDSA with SIC, Frame Slotted ALOHA and IRSA without SIC ($N = 200$ slots, 800 frames per point) 60
Fig 6.12 Packet loss rate versus offered load for the four schemes ($N = 200$ slots, 800 frames per point) 60

LIST OF TABLES

Table 2.1 Relation between the base thesis and the present project 13

Table 3.1 Summary of technical specifications 25
Table 3.2 Differences between the base work and the present project 25
Table 3.3 Boundary between completed work and future work 27

Table 4.1 Functions of the main hardware components 29
Table 4.2 Pin assignment of the four-pin connector J1 29
Table 4.3 CST simulation parameters 31
Table 4.4 GNU Radio model parameters 32
Table 4.5 Parameters of the standalone IRSA/SIC simulation 33
Table 4.6 Comparison of the signal-level and packet-level simulation methodologies . 34

Table 5.1 Band-pass filter settings of the tag receiver branches 43
Table 5.2 Functions of the standalone IRSA/SIC program 46
Table 5.3 SIC iterations for the example frame of Figure 5.14 49

Table 6.1 Simulated RCS at 220 MHz for the two load conditions 53
Table 6.2 BER and power values obtained in the baseline simulation 57
Table 6.3 BER of the three tags for different noise-power settings 57
Table 6.4 Throughput (packets/slot) of IRSA with SIC and the baseline schemes under identical conditions ($N = 200$, 800 frames per load) 59
Table 6.5 Summary of simulation parameters and key throughput results 61
Table 6.6 Achievement of the project objectives 63

Table A.1 Evidence level of the project elements 71

NOMENCLATURE

AGC	Automatic Gain Control: adjusts the signal gain to maintain a suitable output level.
ALOHA	A random-access protocol in which users transmit without a central scheduler.
BER	Bit Error Rate: ratio of incorrectly detected bits to the total number of bits compared.
BPF	Band-Pass Filter
CFO	Carrier Frequency Offset
CRDSA	Contention Resolution Diversity Slotted ALOHA: every packet is sent twice and SIC resolves collisions.
CST	Computer Simulation Technology Studio Suite: an electromagnetic simulation environment.
CW	Continuous Wave: an unmodulated carrier used to illuminate a backscatter tag.
FDMA	Frequency-Division Multiple Access
FIR	Finite Impulse Response (filter)
FSA	Frame Slotted ALOHA: time is divided into frames, and each user transmits once per frame.
GNU Radio	An open-source framework for signal processing and software-defined radio.
GPIO	General-Purpose Input/Output
GRC	GNU Radio Companion
IoT	Internet of Things
IRSA	Irregular Repetition Slotted ALOHA: users transmit a random number of packet replicas drawn from a repetition-degree distribution.
LPF	Low-Pass Filter
OOK	On-Off Keying
PCB	Printed Circuit Board
PLR	Packet Loss Rate: fraction of users whose packets remain undecoded at the end of a frame.
RCS	Radar Cross-Section: a measure of how strongly an object scatters incident electromagnetic energy.
RF	Radio Frequency
SA	Slotted ALOHA
SDR	Software-Defined Radio
SIC	Successive Interference Cancellation: removal of the contribution of decoded packets to recover other packets.
SNR	Signal-to-Noise Ratio

TDMA	Time-Division Multiple Access

Symbols

Γ	Reflection coefficient at the antenna terminals
σ	Radar cross-section (m^2 or dBsm)
G	Offered (channel) load, users per slot
T	Throughput, decoded packets per slot
N	Number of slots in a frame
M	Number of users (tags) contending in a frame
$\Lambda(x)$	Repetition-degree distribution polynomial
F_s, R_s	Sampling rate and symbol rate

CHAPTER 1

INTRODUCTION

1.1 Background

1.1.1 Low-Power Wireless Devices and the Cost of Active Transmission

Wireless links now sit at the centre of industrial automation, environmental monitoring, logistics, agriculture and smart infrastructure. The Internet of Things (IoT) has pushed this trend towards very large numbers of small sensing nodes, each of which reports a few bytes of data at a time. For such nodes the radio is usually the dominant consumer of energy. A conventional transmitter needs a local oscillator, a mixer or modulator, a power amplifier and the associated biasing and filtering, and every one of these blocks draws current whenever a packet is sent. When a deployment contains hundreds or thousands of nodes, the battery replacement effort and the hardware cost of these active chains become the limiting factors rather than the sensing itself [1, 7].

Backscatter communication removes most of this burden. A backscatter device, normally called a *tag*, does not synthesise a carrier at all. It receives a wave that is already present, either from a dedicated emitter or from an ambient source such as a broadcast station, and changes the way its antenna re-radiates that wave. The information is therefore carried by a controlled variation of the reflected field. The tag needs only an antenna, a switching element and a small amount of logic to drive the switch, which allows power consumption in the microwatt range and very low component cost [1, 8].

1.1.2 Principle of Backscatter Communication

Any antenna scatters part of the energy that falls on it. How much is scattered, and with what phase, depends on the impedance connected to the antenna terminals. If the load Z_i is changed, the reflection coefficient

$$\Gamma_i = \frac{Z_i - Z_a^*}{Z_i + Z_a} \tag{1.1}$$

changes with it, where Z_a is the antenna impedance and $(\cdot)^*$ denotes the complex conjugate [1, 10]. A tag that switches its antenna between two loads Z_1 and Z_2 according to a bit stream therefore produces two distinct reflection states, Γ_1 and Γ_2. The reader observes the superposition of the direct illumination, reflections from the environment and the small tag-modulated

component, and it must extract the data from the difference between the two tag states (Figure 1.1).

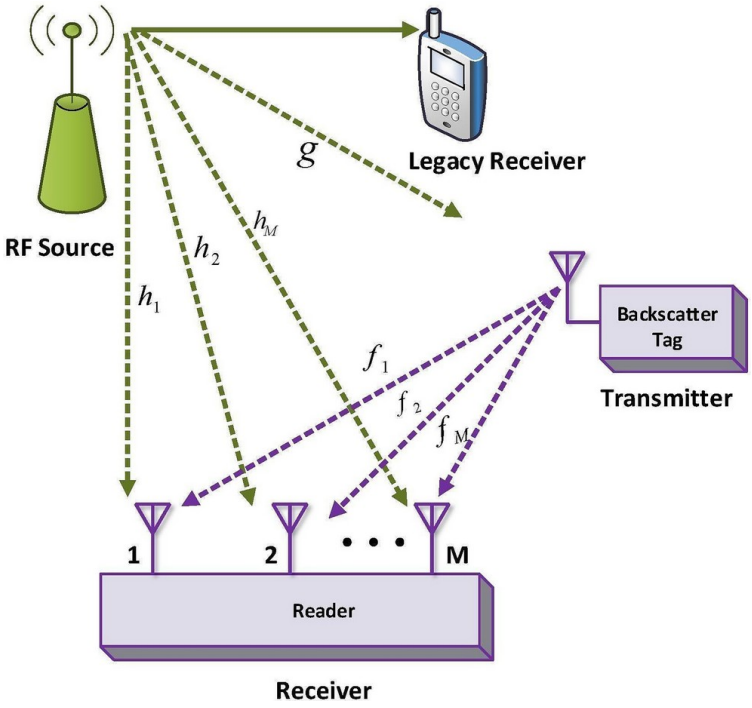


Figure 1.1 – Conceptual overview of a backscatter communication link

Figure 1.2 contrasts this arrangement with conventional active transmission. In the active case the device itself generates, modulates and radiates the RF signal. In the backscatter case the RF energy comes from outside the device, and the tag contributes only the modulation. The penalty for this saving is a weak received signal: the backscattered wave travels the path from the source to the tag and then the path from the tag to the reader, so it suffers two sets of propagation losses and is often far weaker than the direct carrier.

Three architectures are common. In a *monostatic* system, as in passive UHF RFID, the emitter and the receiver are co-located in one reader. In a *bistatic* system the carrier emitter and the reader are physically separate, which allows longer tag-to-reader distances and simpler emitters [9]. In an *ambient* system there is no dedicated emitter at all, and the tag modulates signals that already exist in the environment, such as television, FM radio or Wi-Fi transmissions [6, 2]. The base work of this project, the doctoral thesis of Daskalakis [1], covers both bistatic and ambient FM systems.

When the switching control is itself a square wave at frequency F_{tag}, the reflected spectrum contains components at $F_c \pm F_{tag}$, where F_c is the carrier frequency. These *subcarriers* move the tag information away from the strong carrier and its low-frequency noise. Data can then be imposed on the subcarrier, for example by on-off keying (OOK), in which the subcarrier is

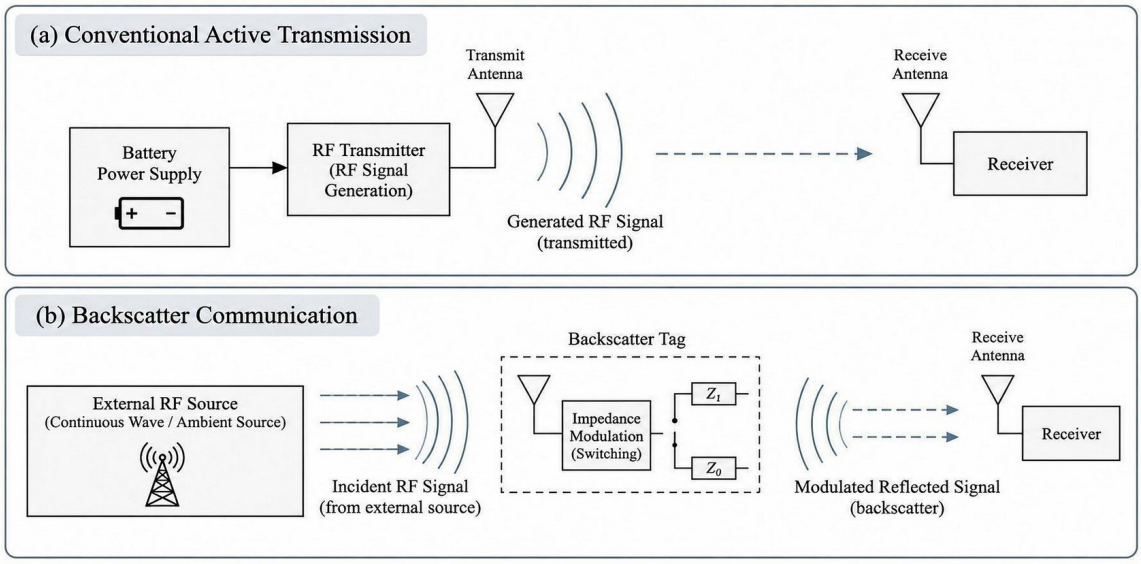


Figure 1.2 – Comparison between conventional active transmission and backscatter communication

either present or absent during each symbol. The GNU Radio model developed in this project is based on exactly this subcarrier mechanism.

1.1.3 Multiple Access in Communication Systems

A multiple-access technique decides how several devices share a common channel. The classical options divide the channel along one physical dimension:

- **Frequency-Division Multiple Access (FDMA)** gives each device its own frequency band. In backscatter systems this is achieved naturally by assigning each tag a different subcarrier frequency F_{tag}.
- **Time-Division Multiple Access (TDMA)** gives each device its own time slots. It requires synchronisation and a schedule that every device follows.
- **Code-Division Multiple Access (CDMA)** lets devices transmit at the same time and frequency with different spreading codes, at the cost of more complex transmitters and receivers.
- **Random access** lets devices transmit whenever they have data, without a central schedule. ALOHA and its slotted and framed variants belong to this family.

Coordinated schemes such as FDMA and TDMA work well when the set of devices is small and known in advance. Backscatter tags, however, are usually very simple, numerous and only intermittently active. They cannot sense the channel before transmitting, and they cannot afford the signalling needed to negotiate a schedule. For this reason random access, in which collisions are tolerated and resolved, is the natural approach for large tag populations. Slotted ALOHA

[15] and Frame Slotted ALOHA [16] are the starting points, and the EPC Class-1 Generation-2 RFID standard uses a framed ALOHA procedure for tag inventory [17].

The weakness of plain random access is that a slot holding two or more packets is simply lost. The best throughput of slotted ALOHA is $1/e \approx 0.37$ packets per slot, so about two thirds of the channel time is wasted on collisions and idle slots. Modern random-access schemes recover part of this loss. In Contention Resolution Diversity Slotted ALOHA (CRDSA) every user sends two copies of its packet [18]; in Irregular Repetition Slotted ALOHA (IRSA) the number of copies is random and follows an optimised distribution [19]. In both schemes the receiver applies Successive Interference Cancellation (SIC): once a packet is decoded from a collision-free slot, its copies are removed from the other slots in which they appear, and this may free further packets for decoding. With a well-chosen repetition distribution, IRSA reaches throughputs far above $1/e$.

1.1.4 Multi-Tag Backscatter and the Base Work

The thesis of Daskalakis [1] demonstrates low-cost, low-power backscatter tags built from a microcontroller, an ADG902 RF switch and a dipole antenna, read by a low-cost software-defined radio. For several tags, the thesis proposes giving each tag its own subcarrier, so that the tags within a cell share the channel by FDMA, and it identifies time-domain multiple access of tags as future work. The present project builds on that foundation. It adopts the switch-loaded tag concept and the subcarrier-based receiver chain, models three tags on separate subcarriers in GNU Radio, and studies IRSA with SIC as the packet-level access method for large numbers of tags.

1.2 Problem Statement

A backscatter reader receives a weak tag-modulated component buried under the direct carrier, environmental reflections and noise. The tag must be designed so that its two load states give a clearly distinguishable scattering response, and the receiver must apply suitable filtering, detection and synchronisation to recover the data. When several tags share the same reader, a second problem appears: their packets overlap, and the reader cannot decode a slot that holds more than one packet. Simple random access then wastes most of the channel time on collisions.

The problem addressed in this project is therefore twofold. At the physical level, it is to design a switch-loaded tag, to verify through electromagnetic simulation that its two states produce a usable scattering contrast, and to model the signal chain of a multi-tag link up to bit recovery. At the access level, it is to implement and evaluate a collision-resolution technique, IRSA with

SIC, that can raise the number of packets recovered per slot well above the limit of slotted ALOHA.

1.3 Motivation

The first motivation is the demand for wireless sensing nodes that need minimal hardware and consume very little power. Equipping every node in a dense sensor network with an active radio is costly in both components and energy, whereas backscatter conveys information by reflection alone.

The second motivation is to understand how circuit design determines communication performance. The load seen by the tag antenna sets its reflection properties, and those properties decide how easily the reader can separate the two states. Combining PCB design with electromagnetic simulation and signal-level modelling makes this relationship concrete.

The third motivation is scalability. Daskalakis notes that multiple access of tags in the time domain is an open problem for the proposed systems [1]. Subcarrier-based FDMA is limited by the number of subcarriers that a tag can produce and a reader can separate, while plain ALOHA wastes most of the channel. IRSA with SIC offers a route to a much higher channel efficiency without requiring the tags to coordinate, which suits simple backscatter devices well.

1.4 Aim of the Project

The aim of the project is to design and model a switch-loaded backscatter communication system through tag hardware design, electromagnetic analysis and signal-level simulation, and to study and independently simulate advanced multiple-access techniques, namely IRSA with SIC, for multi-tag packet recovery.

1.5 Objectives of the Project

1. To study the operating principles of backscatter communication, in particular the relationship between antenna loading, reflection characteristics, subcarrier generation and information representation.
2. To design a compact switch-loaded tag PCB in KiCad, including the power regulation, logic-level translation, RF switch and the four-pin interface to an external ESP32-S3 controller.

3. To analyse the scattering behaviour of the tag in CST Studio Suite and quantify the RCS contrast between its ON and OFF load conditions.
4. To develop a three-tag backscatter signal model in GNU Radio Companion, covering signal generation, subcarrier modulation, noise, receiver processing and BER evaluation.
5. To study IRSA and SIC, implement them independently in Python as a packet-level simulation, and evaluate their throughput and packet loss rate against baseline random-access schemes under identical conditions. Integration of this packet-level work with the developed backscatter system is identified as future work.

1.6 Scope of the Project

The project comprises four workstreams, each addressing a different aspect of multi-tag backscatter communication:

1. **KiCad-based tag hardware design:** schematic capture and PCB layout of a switch-loaded tag with an ADG902 RF switch, a 2.5 V regulator, a level translator and a four-pin connector for the external controller.
2. **CST electromagnetic analysis:** simulation of the tag's radar cross-section for the two switch load conditions over 120–280 MHz.
3. **GNU Radio signal-level modelling:** a three-tag subcarrier model with a complete receiver and BER evaluation for each tag.
4. **Standalone IRSA/SIC implementation in Python:** a Monte Carlo packet-level simulation of IRSA with iterative SIC, together with Frame Slotted ALOHA, CRDSA and a no-SIC reference evaluated under the same conditions.

These workstreams were developed in parallel and were not integrated into one complete physical communication system. The PCB was designed and analysed but not fabricated within the project period, so over-the-air measurements are part of the future scope.

1.7 Overview of the Methodology

Figure 1.3 summarises the approach. A study of backscatter principles and multiple-access techniques, centred on the base thesis, provides the foundation. Four parallel workstreams follow: tag hardware design, electromagnetic analysis, signal-level modelling and packet-level simulation. Their outputs are evaluated individually, and the integration of the four workstreams into one validated system forms the future work.

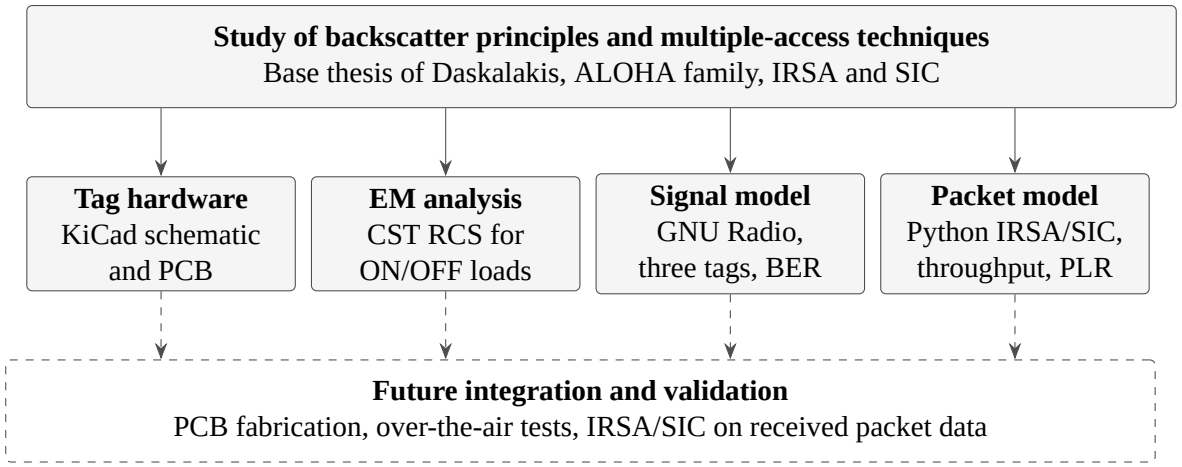


Figure 1.3 – Overall project approach

1.8 Organisation of the Report

Chapter 2 reviews the literature on backscatter communication and multiple access, with a detailed review of the base thesis and its relation to this project. Chapter 3 describes the system architecture, the requirements, the technical specifications and the boundary between completed and future work. Chapter 4 presents the design and methodology of each workstream, including the simulation parameters and assumptions. Chapter 5 details the implementation of the tag design, the CST model, the GNU Radio flowgraph and the standalone IRSA/SIC simulation. Chapter 6 presents and discusses the results. Chapter 7 concludes the report and outlines the future scope. The references and appendices follow Chapter 7.

CHAPTER 2

LITERATURE REVIEW

Backscatter communication has attracted wide attention as a means of connecting devices that cannot afford a conventional active radio. Research in the area spans the electromagnetic behaviour of tags, load modulation circuits, receiver architectures, ambient RF sources and, increasingly, the protocols that let many tags share one reader. This chapter reviews that literature in the order in which it is used in the project. It begins with the fundamentals of backscatter, then reviews the base thesis of Daskalakis [1] in detail and relates it to the present work. It continues with load modulation, radar cross-section, the backscatter link budget and receiver processing, and closes with the multiple-access techniques, from Slotted ALOHA to IRSA with SIC, that form the basis of the packet-level study.

2.1 Evolution of Low-Power Wireless Communication

Radio frequency identification (RFID) was the first large-scale application of the backscatter principle. A passive UHF tag harvests energy from the reader's continuous wave and replies by switching the load on its antenna [12]. Later work extended the idea to semi-passive tags with a small energy store, to bistatic layouts with separate emitters and readers [9], and to ambient backscatter, in which existing television, FM or Wi-Fi signals replace the dedicated emitter [6, 7]. Ambient backscatter removes the carrier source from the system, but it introduces an illumination that is uncontrolled, time-varying and much stronger than the tag signal.

2.2 Fundamentals of Backscatter Communication

A backscatter system contains three functional elements: an RF source that provides the incident wave, a tag that modulates its reflection, and a reader that detects the modulation. The tag's antenna interacts with the incident field, and the impedance at its terminals sets the amplitude and phase of the scattered wave. By switching that impedance in step with a data sequence, the tag imprints the data on the scattered field.

Boyer and Roy [8] analyse coding and energy trade-offs in backscatter links and point out that the tag's choice of modulation and coding directly affects both the reader's error rate and the

energy that a passive tag can harvest. Van Huynh *et al.* [7] survey the ambient backscatter literature and identify interference from the ambient source, the weak modulated signal and multi-tag access as the main open problems. These two works frame the questions that the base thesis and the present project address.

2.3 Review of the Base Work: Ambient Backscatterers for Low-Cost and Low-Power Wireless Applications

The doctoral thesis of S. N. Daskalakis [1], submitted to Heriot-Watt University in 2020, is the main reference for this project. It presents four tag implementations for agricultural and general IoT sensing: a bistatic Morse-coded leaf sensor at 868 MHz [3], an IQ impedance modulator that backscatters LoRa-compatible chirps [5], a binary ambient FM backscatter tag [2], and a four-level pulse amplitude modulation (4-PAM) ambient FM tag [4]. The aspects of the thesis that bear most directly on this project are reviewed below.

2.3.1 Backscatter Principle and Received-Signal Model

The thesis models a bistatic link in which a continuous-wave emitter illuminates the tag and a separate reader receives both the direct carrier and the tag reflection (Figure 2.1). The tag switches its antenna load between two values, producing the reflection coefficients Γ_1 and Γ_2 defined by Equation (1.1). The thesis observes that a phase difference of about 180° between the two reflection coefficients maximises the backscatter performance.

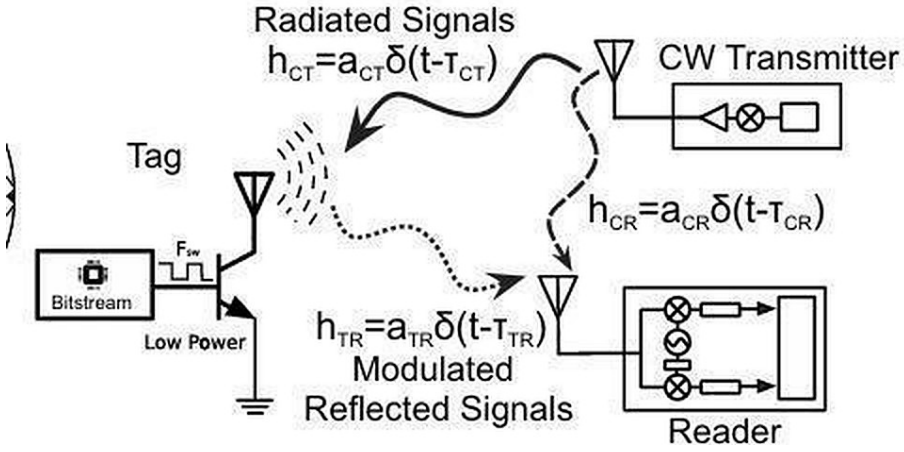


Figure 2.1 – Bistatic backscatter principle with emitter, tag and reader channels. Source: Daskalakis[1].

Following Kimionis *et al.* [9], the complex baseband signal at the reader is written as

$$y_r(t) = n(t) + \frac{A_c}{2} e^{-j2\pi\Delta F t} \left[\alpha_{CR} e^{-j\phi_{CR}} + s\,\alpha_{CT}\,\alpha_{TR} e^{-j\phi_{CTR}}\,\Gamma(t-\tau_{TR}) \right], \tag{2.1}$$

where A_c is the carrier amplitude, ΔF the carrier frequency offset (CFO) between emitter and reader, $\alpha_{CR}e^{-j\phi_{CR}}$ the direct emitter-to-reader channel, $\alpha_{CT}\alpha_{TR}e^{-j\phi_{CTR}}$ the cascaded emitter-to-tag and tag-to-reader channel, s a factor that accounts for the scattering efficiency and gain of the tag antenna, τ_{TR} the tag-to-reader delay and $n(t)$ complex Gaussian noise. Equation (2.1) makes two points clear. First, the information $\Gamma(t)$ appears only in the second term, which is scaled by the product of two channel gains and is therefore much weaker than the direct term. Second, the direct term is a constant offset during a packet, which a receiver can remove as a DC component.

2.3.2 Subcarrier Generation and Frequency-Division Access

In the Morse-coded tag of the thesis, a low-power timer (TS3002) produces a square wave at frequency F_{tag}, and this square wave drives the RF switch. When the incident carrier has frequency F_c, the switching generates two subcarriers at $F_c \pm F_{tag}$ [1]. The microcontroller gates the subcarrier on and off through an ADG902 switch to form the dots and dashes of the Morse code, which is a form of OOK. Figure 2.2 illustrates the resulting spectrum for two tags with different switching frequencies.

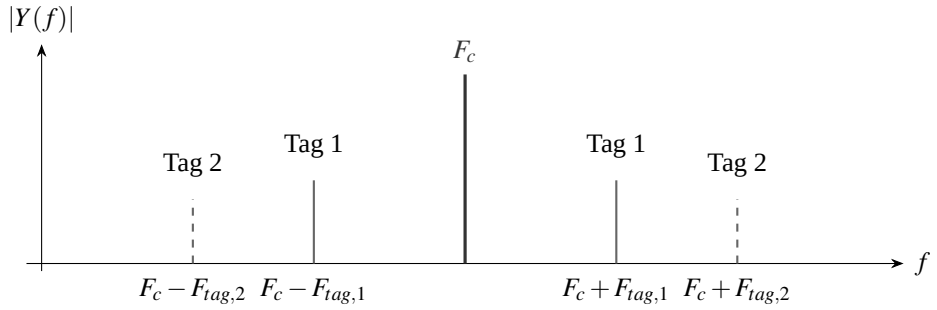


Figure 2.2 – Subcarriers produced by two tags switching at different frequencies around a carrier F_c

The thesis uses this property for multiple access. Each tag in a network can operate at its own F_{tag}; with a digital-to-analogue converter driving the timer's control voltage, the prototype could in principle select one of 32 subcarriers, allowing 32 tags in one network. For larger deployments the thesis proposes cells of tags, in which the tags inside a cell use FDMA on separate subcarriers while the emitters and readers of neighbouring cells share time by TDMA. This subcarrier-based FDMA is the principle on which the three-tag GNU Radio model of this project is built: the three tags switch at 1 kHz, 800 Hz and 1.2 kHz, and the receiver separates them by band-pass filtering around their subcarriers.

2.3.3 Tag Architecture

Across its prototypes the thesis follows a consistent tag architecture: a low-power microcontroller, an RF switch and a dipole antenna. In the ambient FM tag, an MSP430 development board generates the bit stream, and a separate RF front-end board carries the ADG902 switch connected between the two arms of a dipole antenna. A digital output pin of the microcontroller drives the switch control input, and the front-end draws its supply from the development board. The ADG902 was chosen for its low insertion loss (about 0.5 dB at 100 MHz), high OFF isolation (about 57.5 dB at 100 MHz) and current consumption below 1 μA. Its dynamic power follows $\frac{1}{2}C_{RF}V_{DD}^2F_{sw}$, which the thesis estimates at only 16.3 nA for a 2.5 kbps switching rate.

The tag designed in this project follows the same separation of functions. The KiCad board contains the ADG902 switch, its 2.5 V regulator, a level translator and the antenna pads, while the ESP32-S3 controller sits on a separate board and connects to the tag through a four-pin connector carrying the supplies, the control signal and ground. This modular arrangement mirrors the development-board-plus-front-end structure of the base work and allows the RF board to be tested independently of the controller.

2.3.4 Receiver Processing Chain

The thesis uses a low-cost RTL-SDR as the reader, connected through GNU Radio to a MATLAB decoding algorithm [1]. For the Morse-coded tag the receiver performs CFO correction, takes the absolute value of the signal, applies a band-pass filter centred on F_{tag}, takes the magnitude again, applies a matched filter whose length equals one symbol, downsamples, and finally applies an AGC and a threshold to decide each symbol. For the ambient FM tag, the receiver forms the squared magnitude $|y_r[k]|^2$, which removes the CFO without explicit estimation, then applies a matched filter, removes the DC offset estimated over a window without tag activity, corrects for waveform inversion, and synchronises to the packet by cross-correlating with a known preamble.

The receiver of the present GNU Radio model follows the same sequence of operations: band-pass filtering around each tag's subcarrier, envelope detection with an absolute-value block, low-pass smoothing, DC removal, a matched (moving-average) filter one symbol long, AGC, symbol synchronisation and a binary slicer. The correspondence between the two chains is summarised in Table 2.1.

2.3.5 Ambient FM Backscatter and Error Analysis

The binary ambient FM tag of the thesis uses OOK with FM0 encoding, the line code of passive RFID. Each FM0 bit contains two symbols with a phase inversion at every bit boundary and an extra inversion in the middle of a “0”, and a dummy “1” terminates each packet. Packets of 26 bits carry a 10-bit preamble, a 2-bit tag identifier, a 2-bit sensor identifier and the sensor data [1]. The thesis derives the bit error probability by noting that, after a one-symbol shift, FM0 reduces to an antipodal constellation. If P_s is the error probability of the shifted signal, the bit error probability after differential decoding is

$$P_e = 2P_s(1 - P_s), \tag{2.2}$$

and the antipodal view yields an SNR advantage of about 3 dB over detection based on the bi-orthogonal constellation. Measurements in an anechoic chamber agreed well with this analysis. In an indoor deployment using an FM station 34 km away, the tag communicated over a 5 m tag-to-reader distance at bit rates of 0.5 to 2.5 kbps, with an energy cost of 36.9 μJ per packet at 2.5 kbps.

2.3.6 High-Order Modulation and Future Directions

The 4-PAM ambient FM tag replaces the switch with a single transistor whose gate voltage, set by a DAC, selects one of four reflection coefficients. This doubles the bit rate for the same symbol rate, at a power consumption of 27 μW and an energy of 78.2 nJ/bit at 345 bps [4]. In discussing larger networks, the thesis notes that multiple tags could share one reader through a TDMA scheme and that tags could transmit at random intervals to avoid collisions. It explicitly lists multiple access of tags in the time domain as future work.

2.3.7 Relation of the Base Work to the Present Project

Table 2.1 maps the main elements of the base thesis to the corresponding work in this project. The project adopts the switch-loaded tag, the subcarrier principle and the receiver structure of the thesis, and it extends the multi-tag question that the thesis leaves open by studying a random-access scheme with collision resolution.

Table 2.1 – Relation between the base thesis and the present project

Aspect	Base thesis [1]	Present project
Tag front-end	ADG902 switch between dipole arms, driven by an MCU pin	ADG902 switch on a KiCad tag PCB, driven through a level translator
Controller	MSP430 or PIC MCU, separate from the RF board	ESP32-S3 on a separate board, connected through a four-pin connector
State contrast	Two-state Γ on a Smith chart; measured links	Simulated RCS for ON (2 Ω) and OFF (0.45 pF) loads in CST
Multi-tag access	Subcarrier FDMA (up to 32 subcarriers); TDMA proposed	Three subcarriers in GNU Radio; IRSA with SIC at packet level
Receiver	SDR with MATLAB: BPF, envelope, matched filter, AGC, threshold	GNU Radio: BPF, Abs, LPF, DC block, matched FIR, AGC, symbol sync, slicer
Performance metric	Measured BER and PER versus distance and power	Simulated BER versus SNR; packet-level throughput and PLR

2.4 Load Modulation and Reflection Coefficient

Load modulation varies the impedance connected to the antenna so that its reflection follows the information. With two impedance states, each bit period selects one of two reflection coefficients from Equation (1.1); with more states, as in the 4-PAM tag of the base work, several bits can be sent per symbol. The modulation depth seen by the reader depends on the difference $|\Gamma_1 - \Gamma_2|$: the further apart the two states lie on the Smith chart, the larger the change in the scattered field [10].

In practice the reflection coefficient cannot be separated from the antenna and its surroundings. The switch adds its own on-resistance and off-capacitance, the PCB traces add parasitics, and nearby objects detune the antenna, so the effective states can differ from the ideal short and open. This is why the project models the switch states in CST as a small resistance (2 Ω) for the ON state and a small capacitance (0.45 pF) for the OFF state rather than as an ideal short and open circuit.

Figure 2.3 illustrates the two-state switching used throughout the project. In the GNU Radio model the same idea appears as two numerical response coefficients, σ_{ON} and σ_{OFF}, selected by the control waveform.

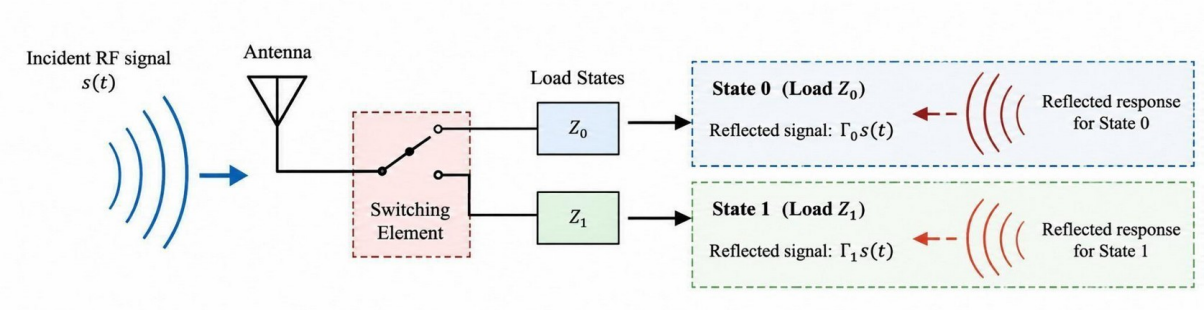


Figure 2.3 – Two-state load switching at a backscatter tag

2.5 Antenna Behaviour and Radar Cross-Section

Radar cross-section (RCS) expresses how strongly an object scatters incident power in a specified direction. It is defined as the area that would intercept enough power from the incident wave to produce, if radiated isotropically, the observed scattered power density. For an antenna terminated in a load, the RCS can be written as [10]

$$\sigma = \frac{\lambda^2 G^2}{4\pi} |A_s - \Gamma|^2, \tag{2.3}$$

where λ is the wavelength, G the antenna gain and A_s a load-independent term related to the structural scattering of the antenna. Equation (2.3) explains directly why switching the load changes the scattered power, and why the difference between the RCS values of the two states is a convenient single measure of how distinguishable they are. RCS also depends on frequency, polarisation and observation angle, so any RCS result applies to the specified geometry and illumination.

2.6 Backscatter Link Budget

The power reaching the reader from the tag follows the bistatic radar equation [11]:

$$P_R = \frac{P_T G_T G_R \lambda^2 \sigma}{(4\pi)^3 d_1^2 d_2^2}, \tag{2.4}$$

where P_T is the transmitted power, G_T and G_R the emitter and reader antenna gains, d_1 the emitter-to-tag distance and d_2 the tag-to-reader distance. The received tag power falls as the product $d_1^2 d_2^2$, which is why backscatter ranges are short compared with one-way links of the same power. Griffin and Durgin [11] extend this equation with polarisation mismatch, on-object gain penalties and fading margins to form complete link budgets.

For a fixed geometry, Equation (2.4) makes the received power proportional to σ. The ratio of the received powers in the two tag states is therefore equal to the ratio of their RCS values, so an

RCS contrast expressed in decibels translates directly into the same contrast in received power. This is the link between the CST results of this project and the modulation depth available at a reader.

2.7 Receiver Processing in Backscatter Systems

The receiver must detect a small modulation in the presence of the carrier, reflections, interference and noise. Typical processing includes CFO correction or squaring to remove it, filtering to isolate the tag band, envelope detection, matched filtering to maximise the SNR at the sampling instants, DC removal, gain control, timing synchronisation and a decision device [1, 13]. GNU Radio provides these operations as connectable blocks and allows a complete chain to be assembled and tested in software before any RF hardware is used [21]. The base thesis already uses GNU Radio as the interface between its SDR and its decoder, and this project uses GNU Radio Companion for the whole signal-level model.

2.8 Challenges in Multi-Tag Communication

When several tags respond to the same reader, their reflections overlap, and a reader that receives two packets in the same slot usually decodes neither. Subcarrier FDMA avoids this for a small number of tags, but the number of usable subcarriers is limited by the tag oscillator, the receiver bandwidth and the filter selectivity, and every tag must be assigned a frequency in advance.

For large or changing tag populations, random access is more practical, since the tags need no coordination. The price is collisions. Three general strategies reduce their impact: organising transmissions into slots and frames so that the probability of overlap is controlled, sending several copies of each packet so that at least one copy is likely to be received cleanly, and using decoded packets to cancel interference in the slots where they collided. IRSA combines the second and third strategies within the framework of the first.

2.9 Multiple-Access Techniques

Multiple-access methods fall into two broad classes. Scheduled methods such as FDMA, TDMA and CDMA assign each user a dedicated resource; they avoid collisions but require coordination. Random-access methods let users transmit without coordination and resolve, or tolerate, collisions. The access protocol and the backscatter mechanism perform different functions. Load switching determines how a tag encodes bits in the reflected wave, the access protocol determines when each tag transmits its packets, and SIC is a receiver-side procedure that operates on the resulting slot observations. The following sections review the random-access protocols relevant to this project in order of increasing capability.

2.10 Slotted ALOHA

Pure ALOHA, introduced by Abramson for the ALOHAnet radio network [14], lets users transmit whenever they have data. Roberts [15] divided time into slots and required transmissions to start at slot boundaries, which halves the vulnerable period. If packets arrive as a Poisson process with an average of G packets per slot, a slot carries a successful packet only when exactly one packet is present, so the throughput is

$$S_{SA} = G e^{-G}. \tag{2.5}$$

The maximum, $1/e \approx 0.368$, occurs at $G = 1$. Figure 2.4 illustrates the principle.

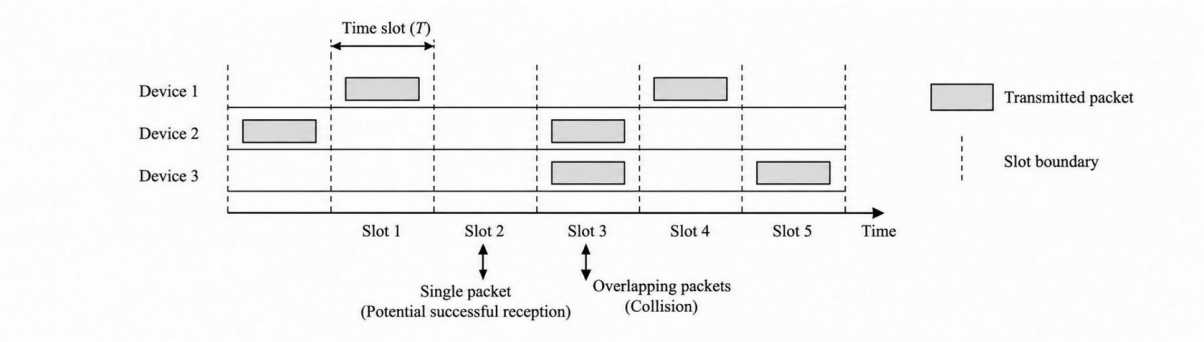


Figure 2.4 – Slotted ALOHA access principle

Slotted ALOHA is simple because no per-packet scheduling is needed, but it cannot recover a packet from a collided slot and makes no use of information from other slots. These limitations motivated the replica-based methods reviewed below.

2.11 Frame Slotted ALOHA

In Frame Slotted ALOHA (FSA) the slots are grouped into frames of N slots, and each of the M active users transmits its packet once per frame in a slot chosen uniformly at random (Figure 2.5). A user is successful when no other user picks the same slot, so the expected throughput in packets per slot is

$$T_{FSA} = \frac{M}{N}\left(1-\frac{1}{N}\right)^{M-1} \approx G e^{-G}, \qquad G = \frac{M}{N}, \tag{2.6}$$

where the approximation holds for large N. Schoute [16] showed that adapting the frame length to the estimated number of users keeps the system near its optimum, and the EPC Class-1 Generation-2 RFID protocol uses an adaptive framed ALOHA procedure (the Q algorithm) for tag inventory [17].

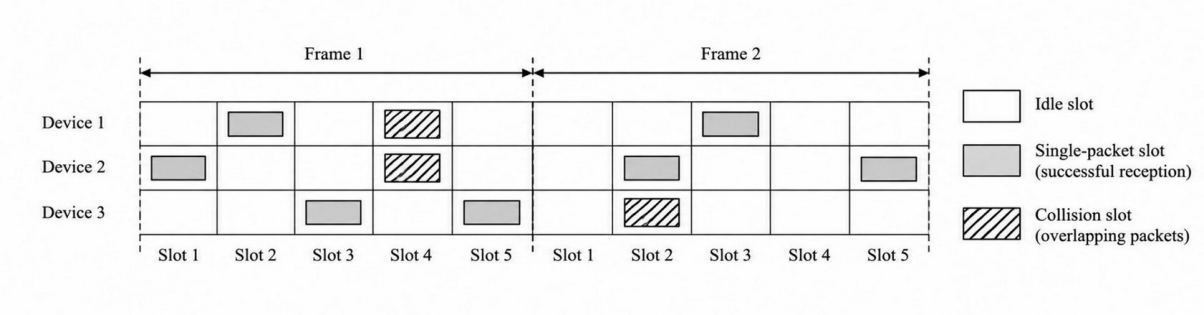


Figure 2.5 – Frame Slotted ALOHA access principle

FSA is the natural baseline for the IRSA study in this project, because IRSA keeps the same frame structure and changes only the number of copies each user sends and the way the receiver processes collided slots.

2.12 Contention Resolution Diversity Slotted ALOHA

Casini *et al.* [18] proposed CRDSA for satellite random access. Each user transmits two copies of its packet in two different slots of a frame, and each copy carries a pointer to the slot of its twin. When one copy is received in a clean slot and decoded, the receiver knows where the other copy lies, reconstructs its waveform and subtracts it from that slot. The cancellation may leave another packet alone in that slot, which can then be decoded in turn. This iterative procedure is the SIC process used throughout this report. CRDSA raises the peak throughput of slotted ALOHA from about 0.37 to about 0.55 packets per slot.

2.13 Irregular Repetition Slotted ALOHA

Liva [19] generalised CRDSA by letting the number of copies vary from user to user. Before each frame, a user draws its repetition degree ℓ with probability Λ_ℓ and transmits ℓ copies in distinct slots chosen at random. The distribution is written compactly as a polynomial,

$$\Lambda(x) = \sum_{\ell} \Lambda_\ell x^\ell, \qquad \sum_{\ell} \Lambda_\ell = 1, \tag{2.7}$$

and the average number of copies per user is $\bar{\ell} = \Lambda'(1) = \sum_\ell \ell \Lambda_\ell$. The word "irregular" refers to this variation of the degree among users. Figure 2.6 illustrates the transmission of replicas within a frame.

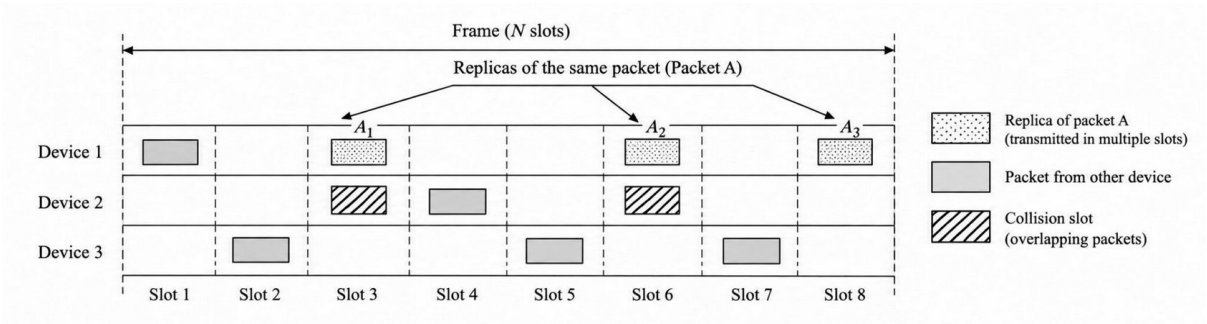


Figure 2.6 – IRSA replica-transmission concept

Liva's central observation is that the SIC process is equivalent to iterative erasure decoding on a bipartite graph. Users form one set of nodes, slots form the other, and an edge joins a user to every slot that holds one of its copies. Decoding a user from a slot of degree one and cancelling its other copies is the same as removing that user node and all its edges. This equivalence allows the repetition distribution to be optimised with the density-evolution tools developed for low-density parity-check codes. For each distribution there is a load threshold G^*: for large frames, almost all users are recovered when $G < G^*$, and decoding collapses when G exceeds it. Optimised irregular distributions push G^* well above the CRDSA value. The distribution $\Lambda(x) = 0.5x^2 + 0.28x^3 + 0.22x^8$, used in this project, is one of the optimised distributions reported by Liva, with an asymptotic threshold of about 0.94 packets per slot [19].

Two practical effects keep finite systems below this asymptotic behaviour. With a finite number of slots the transition around G^* becomes gradual and moves to lower loads. In addition, small structures in the graph, such as two users that both choose the same two slots, cannot be resolved by SIC at any load; these *stopping sets* produce an error floor in the packet loss rate at low load. Paolini *et al.* [20] developed the broader framework of coded slotted ALOHA, in which packets are encoded with local codes rather than simply repeated, and analysed these finite-length effects in detail.

2.14 Successive Interference Cancellation

SIC is the receiver operation that turns repetition into a throughput gain. In a physical receiver it requires three steps: decoding the packet from a clean slot, reconstructing the waveform of its copies in the other slots (including channel gain, phase and timing), and subtracting those copies from the stored slot signals. Imperfect channel estimation leaves residual interference after subtraction, which reduces the benefit of SIC. In packet-level analysis, the common assumption is ideal cancellation on a collision channel: a slot holding exactly one packet is decoded, a slot holding more than one is not, and cancellation removes a copy completely [18, 19]. The packet-level simulation in this project adopts this model, which isolates the effect of the access protocol from the physical-layer details studied in the GNU Radio model.

2.15 Performance Evaluation Metrics

Two families of metrics are relevant. At the packet level, the *throughput* T is the average number of successfully decoded packets per slot, and the *packet loss rate* (PLR) is the fraction of users whose packet remains undecoded at the end of a frame. The offered load G is the number of users divided by the number of slots. With these definitions, an ideal access scheme with no losses has $T = G$. At the physical level, the relevant metrics are the contrast between tag states (here expressed as the RCS difference), the SNR at the receiver and the bit error rate (BER). Physical-layer metrics describe the quality of an individual link, while packet-level metrics describe how efficiently many tags share the channel. The two kinds of results are therefore reported separately in this project.

2.16 Research Gap

The reviewed literature establishes two mature but largely separate bodies of work. Backscatter research, including the base thesis, has produced low-power tags and practical receivers, and it handles multiple tags mainly by assigning separate subcarriers, leaving time-domain multiple access as an open problem [1]. Random-access research has produced IRSA and coded slotted ALOHA, which approach a throughput of one packet per slot, but it has been developed mainly for satellite and terrestrial radio networks with capable transmitters [18, 19, 20].

Multi-tag backscatter needs both: a physical layer that delivers a detectable modulation from each tag, and an access layer that resolves collisions among many uncoordinated tags. A complete study therefore has to look at signal-level processing and packet-level collision resolution together. This project approaches the problem through separate but complementary implemen-

tations. The tag design and its electromagnetic analysis address the physical source of the modulation, the GNU Radio model addresses the separation and detection of several tags at signal level, and the Python simulation quantifies what IRSA with SIC can achieve at packet level under the standard collision-channel assumptions. Bringing these layers together in one experimentally validated system is the step that remains after this work.

CHAPTER 3

SYSTEM ARCHITECTURE

This chapter describes how the project is organised as a system. It identifies the four workstreams and the role of each, presents the architecture diagram, sets out the functional and non-functional requirements, specifies the hardware and software architecture with their inputs and outputs, lists the technical specifications and the differences from the base work, and defines the boundary between the completed work and the future work.

3.1 System Overview

The project studies multi-tag backscatter communication from four complementary angles, organised as four workstreams:

1. **Hardware design.** A switch-loaded backscatter tag is designed in KiCad. The tag PCB carries the ADG902 RF switch, its regulator, a level translator and the antenna pads, and provides a four-pin interface for an external ESP32-S3 controller.
2. **Electromagnetic analysis.** The tag design is imported into CST Studio Suite, and its radar cross-section is computed for the ON and OFF switch load conditions to quantify the contrast between the two backscatter states.
3. **Signal-level modelling.** A GNU Radio Companion flowgraph models three tags on separate subcarriers, combines their responses with the carrier and noise, and recovers each tag's bits in a dedicated receiver branch with BER evaluation.
4. **Packet-level multiple-access simulation.** A standalone Python program implements IRSA with iterative SIC on a slotted collision channel and evaluates throughput and packet loss rate over a range of offered loads, alongside baseline schemes under identical conditions.

Each workstream investigates a different layer of the same communication problem. The hardware design and the CST analysis concern the physical origin of the modulation, the GNU Radio model concerns the separation and detection of several tags at signal level, and the Python simulation concerns how efficiently many tags can share the channel at packet level. The four

workstreams were developed in parallel, with shared parameters and assumptions where relevant, but they are not directly interconnected: the outputs of one are not fed as inputs to another.

3.2 System Architecture Diagram

Figure 3.1 presents the system architecture. The four workstreams appear as parallel blocks, each with its main tool, its principal activity and its output. The common theoretical basis, drawn from the base thesis and the multiple-access literature, sits above them. The dashed outline below the four blocks marks the proposed integration and validation stage, which would connect the fabricated tag, an RF set-up, the signal-level receiver and the packet-level access scheme into one system.

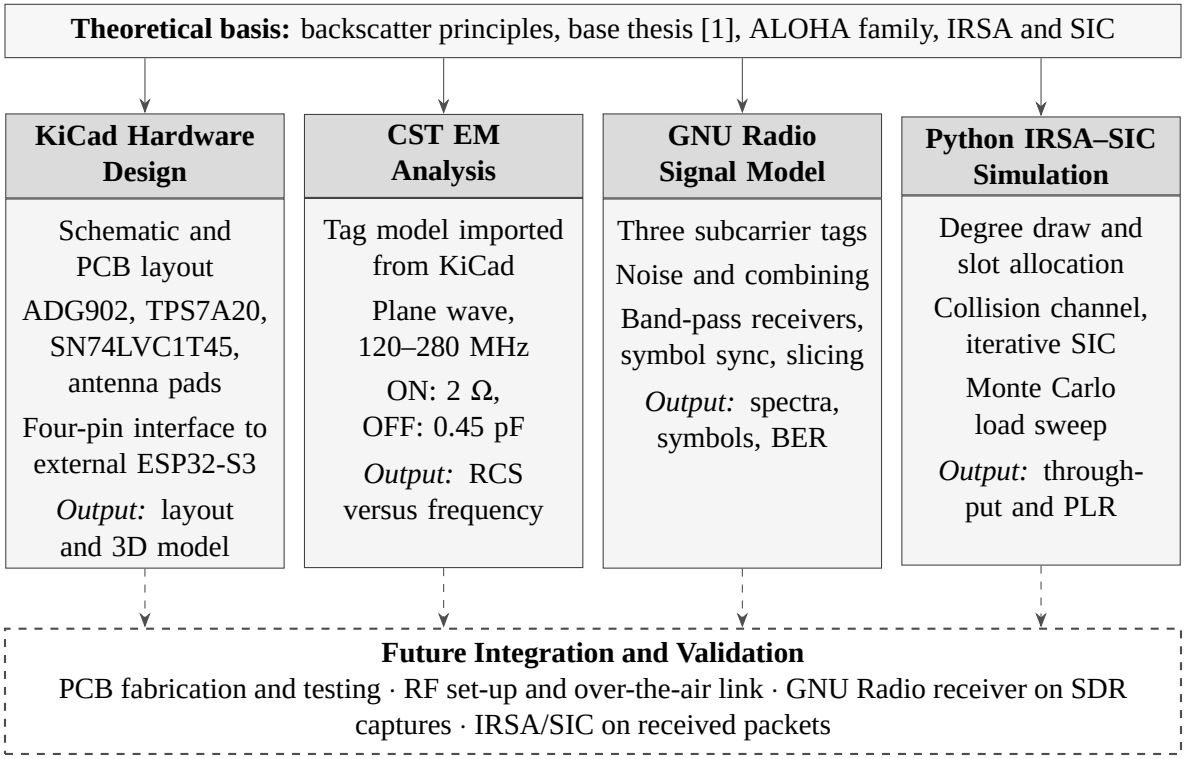


Figure 3.1 – System architecture of the project: four parallel workstreams and the proposed integration stage

The diagram reflects the actual structure of the work. The arrows from the theoretical basis indicate that every workstream draws on the same principles; the absence of arrows between the four blocks indicates that they were executed independently; and the dashed arrows into the lower outline indicate how each block would contribute to a future integrated system. In that system, the fabricated tag would supply the physical modulation whose contrast the CST model predicts, an SDR would capture the reflected signal, a receiver of the kind modelled in GNU

Radio would recover the packets, and the IRSA/SIC procedure would resolve collisions among the packets of many tags.

3.3 Functional Requirements

1. **Tag switching function.** The tag PCB shall accept a digital control signal from an external controller and translate it to the logic level required by the RF switch, so that each control level selects one of two antenna load conditions.
2. **Controller interface.** The tag PCB shall provide a four-pin connector carrying the 5 V supply, the 3.3 V logic supply, the control signal and the common ground from the ESP32-S3 controller board.
3. **Electromagnetic characterisation.** The CST model shall compute the RCS of the tag for both load conditions over a defined frequency range, so that the state contrast can be quantified at the frequency of interest.
4. **Signal-level model.** The GNU Radio model shall generate data, carrier and subcarrier control signals for three tags, model the two-state tag response, add Gaussian noise, combine all signals, and recover the bits of each tag in a separate receiver branch.
5. **BER evaluation.** The signal model shall compare the recovered bits of each tag with a time-aligned reference and compute the BER, and it shall allow the noise level to be varied to obtain BER as a function of SNR.
6. **Packet-level IRSA/SIC.** The Python program shall assign repetition degrees according to a specified distribution, place packet replicas in distinct random slots of a frame, model collisions on a slotted channel, and decode users through iterative SIC.
7. **Throughput evaluation.** The packet-level program shall compute the throughput and PLR of each frame, average them over a defined number of Monte Carlo trials, and sweep the offered load over a specified range. It shall support baseline schemes evaluated under identical frame and trial settings.

Physical RF transmission, fabrication of the tag and integration of the packet-level scheme with the tag hardware or the GNU Radio model are not part of the completed requirements; they belong to the integration stage in Figure 3.1.

3.4 Non-Functional Requirements

Modularity. Each workstream should be self-contained so that it can be examined, modified and extended without redesigning the others. In the GNU Radio model, the three tag branches and their receivers are built from the same block structure; in the Python program, frame generation, decoding and Monte Carlo averaging are separate functions.

Reproducibility. Design files, flowgraph settings, source code and simulation parameters should be retained so that every reported value can be regenerated. The packet-level simulation uses a fixed master seed, so repeated runs produce identical results.

Consistency of comparison. Baseline and proposed access schemes must be evaluated with the same frame size, load points, number of trials and random seed, so that differences in the results arise only from the schemes themselves.

3.5 Hardware Architecture and Requirements

The hardware is split across two boards. The *controller board* carries an ESP32-S3 WROOM module, powered from a 5 V USB supply through an AMS1117-3.3 regulator; one of its GPIO pins generates the switching control signal. The *tag PCB* designed in KiCad carries the RF section: a TPS7A20 low-dropout regulator that derives a 2.5 V rail from the 5 V input, an SN74LVC1T45 level translator that converts the 3.3 V control logic to 2.5 V, the ADG902 RF switch and the pads for the two antenna arms. The two boards are joined by the four-pin connector J1; the controller is external to the tag and is not mounted on the tag PCB. This split follows the base work, in which a microcontroller development board drives a separate RF front-end board [1].

A complete physical link would additionally require an RF source to illuminate the tag, antennas for the source and the reader, and an SDR receiver with a host computer. These items define the equipment for the future validation stage.

3.6 Software Architecture and Requirements

Four software environments are used, one per workstream: KiCad for schematic capture and PCB layout, CST Studio Suite for the electromagnetic simulation, GNU Radio Companion for the signal-level model, and Python 3 with NumPy and Matplotlib for the packet-level simulation and plotting [22, 23].

3.7 Technical Specifications

3.7.1 Key Parameters

Table 3.1 summarises the principal technical parameters of the four workstreams. Detailed parameter tables appear in Chapter 4.

Table 3.1 – Summary of technical specifications

Parameter	Specification
RF switch	ADG902 (reflective SPST), CTRL at 2.5 V logic
Tag supply	5 V input, regulated to 2.5 V by TPS7A20
Controller	ESP32-S3 WROOM on a separate board, 3.3 V GPIO
Controller–tag interface	Four-pin connector J1: +5 V, +3.3 V, CTRL, GND
CST sweep and loads	120–280 MHz; ON 2 Ω, OFF 0.45 pF
GNU Radio sampling and symbol rate	50 kS/s, 20 symbols/s (2500 samples/symbol)
Subcarrier frequencies	1 kHz, 800 Hz and 1.2 kHz on a 10 kHz carrier
IRSA frame size	200 slots
Degree distribution	$\Lambda(x) = 0.5x^2 + 0.28x^3 + 0.22x^8$ (mean 3.6 replicas)
Load range and trials	$G = 0.10$ to 1.00 in steps of 0.05; 800 frames per load

3.7.2 Differences from the Base Work

The project draws on the base thesis but differs from it in several respects, summarised in Table 3.2.

Table 3.2 – Differences between the base work and the present project

Item	Base thesis [1]	This project
Illumination	868 MHz CW emitter; ambient FM (88–108 MHz)	Electromagnetic study at 120–280 MHz; baseband-equivalent 10 kHz carrier in GNU Radio
Switch drive	MCU pin directly at the switch supply level	ESP32-S3 at 3.3 V through a level translator to 2.5 V
Evidence	Fabricated tags; anechoic and indoor measurements	PCB design, CST simulation, GNU Radio and Python simulations
Multi-tag method	Subcarrier FDMA; TDMA suggested	Subcarrier separation at signal level; IRSA with SIC at packet level
Line code	Morse OOK, FM0-coded OOK, 4-PAM	OOK on a subcarrier with repeating test sequences

3.8 Design Constraints and Assumptions

Project time. The project period did not allow PCB fabrication, assembly and RF testing; the hardware stream was therefore carried through design and electromagnetic simulation.

Electromagnetic model. The CST results apply to the modelled geometry, plane-wave excitation, polarisation and load models. Component tolerances and the effect of the controller cable are not included.

Signal model. The GNU Radio model represents the carrier at a baseband-equivalent frequency of 10 kHz, represents the tag states by numerical response coefficients and uses additive Gaussian noise. Multipath, carrier frequency offset and phase noise are not modelled.

Packet model. The IRSA simulation assumes slot-synchronous transmission, one packet per user per frame, a collision channel without capture, knowledge of replica positions at the receiver, and ideal cancellation. These are the standard assumptions of the IRSA literature [18, 19] and are stated again with the simulation parameters in Chapter 4.

3.9 Feasibility Analysis

Technical feasibility. All four workstreams use established tools (KiCad, CST, GNU Radio and Python with NumPy), and the selected components (ADG902, TPS7A20, SN74LVC1T45, ESP32-S3) are widely available and well documented.

Economic feasibility. The design and simulation stages need no material cost, and the tag uses a few low-cost surface-mount parts with an off-the-shelf controller, so a prototype is inexpensive. An SDR and an RF source would be needed for over-the-air testing.

Operational feasibility. Each workstream produces outputs that can be inspected and reproduced independently, and the IRSA/SIC procedure is a sequence of slot-level operations that can later run on packets extracted by an SDR receiver.

3.10 System Boundary

Table 3.3 defines the boundary between the work completed in this project and the work planned for later stages.

Table 3.3 – Boundary between completed work and future work

Completed in this project	**Future work**
Tag schematic, PCB layout and 3D model in KiCad, with a four-pin interface to an external ESP32-S3	Fabrication, assembly and electrical testing of the tag PCB
CST RCS analysis of the tag for ON and OFF load conditions	Measured RCS and over-the-air backscatter link
Three-tag GNU Radio model with receivers, BER evaluation and BER-versus-SNR sweep	Operation of the receiver on SDR captures of real tags
Standalone Python IRSA/SIC simulation with throughput and PLR, and FSA, CRDSA and no-SIC baselines	Integration of IRSA/SIC with the signal-level receiver and the tag hardware; imperfect SIC and capture

CHAPTER 4

SYSTEM DESIGN AND METHODOLOGY

4.1 Design Overview

The design follows the four workstreams defined in Chapter 3. The hardware design produces a switch-controlled RF backscatter tag whose control signal comes from an external ESP32-S3 controller through a four-pin connector (Figure 4.2). The electromagnetic analysis characterises the scattering of the tag in its two load states. The GNU Radio model represents three subcarrier tags, their combination with noise and their separate recovery at the receiver. The packet-level simulation evaluates IRSA with SIC on a slotted collision channel. This chapter sets out the design of each workstream, the tools used, the parameters and assumptions of each simulation, and the metrics used to assess the results.

4.2 Hardware Design

The tag hardware uses an ESP32-S3 WROOM controller, an AMS1117-3.3 regulator, a TPS7A20 regulator, an SN74LVC1T45 level translator and an ADG902 RF switch (Table 4.1) [29, 27, 28, 26]. The ESP32-S3 and the AMS1117-3.3 are on the controller side: the AMS1117-3.3 derives the controller's 3.3 V rail from the 5 V USB supply, and a GPIO pin of the ESP32-S3 provides the control signal. The remaining parts are on the tag PCB. The TPS7A20 regulates the incoming 5 V to 2.5 V for the RF switch and for the output side of the translator, and the SN74LVC1T45 converts the 3.3 V control logic to the 2.5 V level required at the CTRL input of the ADG902. The connector J1 links the two boards, with the pin assignment in Table 4.2.

4.3 Electromagnetic Analysis Design

The tag design was imported from KiCad into CST Studio Suite to study its backscattering properties. The structure is a dipole tag whose feed carries a lumped element representing the RF switch. The ON state is modelled as a 2 Ω resistance, representing the low on-resistance of the conducting switch, and the OFF state as a 0.45 pF capacitance, representing the off-state capacitance of the open switch. The tag is illuminated by an X-polarised plane wave with an electric-field amplitude of 1 V/m, and the RCS is computed over 120–280 MHz for each state.

Table 4.1 – Functions of the main hardware components

No.	Component	Location	Function
1	ESP32-S3 WROOM	Controller board	Generates the digital control signal on a GPIO pin
2	AMS1117-3.3	Controller board	Provides the 3.3 V controller rail from the 5 V USB supply
3	TPS7A20	Tag PCB	Regulates the 5 V input to 2.5 V
4	SN74LVC1T45	Tag PCB	Translates the 3.3 V control logic to 2.5 V
5	ADG902	Tag PCB	RF switch; selects the antenna load state

Table 4.2 – Pin assignment of the four-pin connector J1

Pin	Signal	Description
1	+5 V	Supply input to the tag regulator
2	+3.3 V	Input-side logic supply of the translator
3	CTRL	Digital control signal from an ESP32-S3 GPIO
4	GND	Common ground

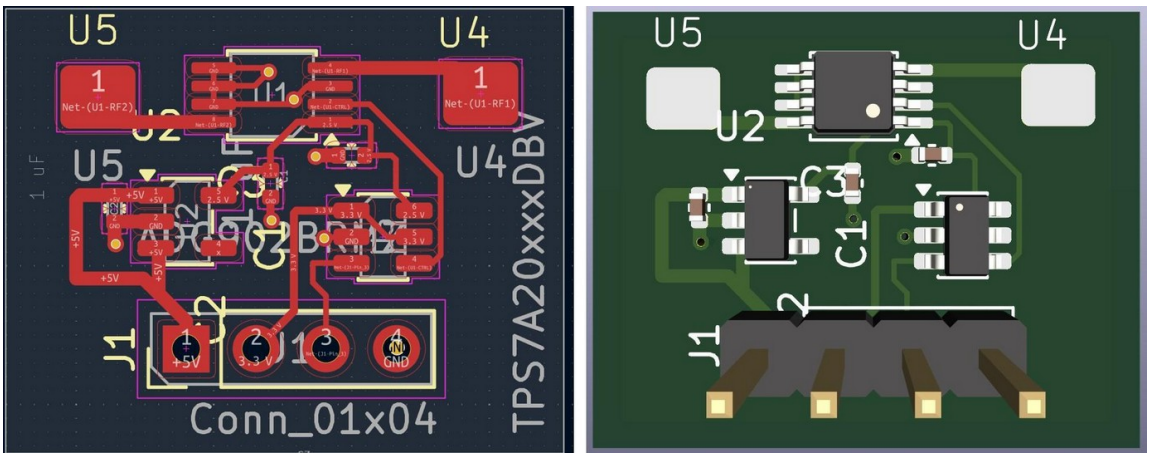


Figure 4.1 – KiCad PCB layout and 3D model of the RF backscatter tag

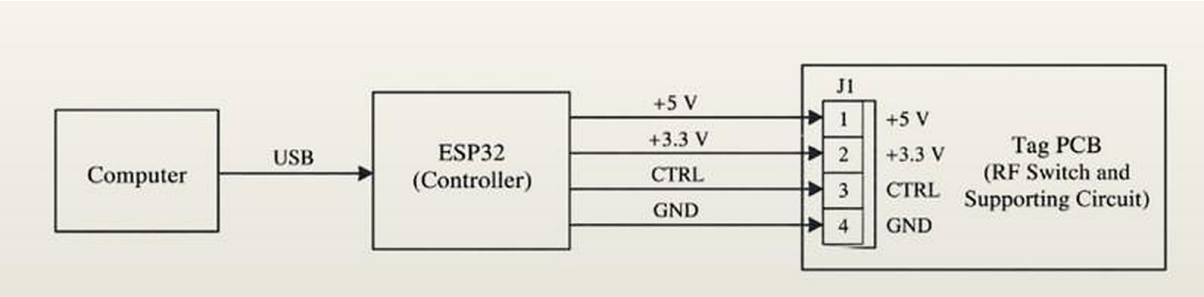


Figure 4.2 – System-level interface between the computer, the external ESP32-S3 controller and the RF tag PCB through connector J1

The figure of merit is the difference between the two RCS curves at the frequency of interest, which by Equation (2.4) equals the difference in received backscatter power between the two states for a fixed geometry.

4.4 Signal-Level Modelling Design

The GNU Radio model represents three tags that share the channel by subcarrier separation, the principle described in Section 2.3.2. Each tag switches a 10 kHz carrier between two response coefficients under the control of its own square-wave subcarrier (1 kHz, 800 Hz or 1.2 kHz) gated by its data. The tag outputs, a carrier term and Gaussian noise are summed into one received signal, and a dedicated receiver branch per tag recovers the bits for BER evaluation. The flowgraph is described block by block in Section 5.4.

4.5 Packet-Level Multiple-Access Design

The packet-level model follows the standard IRSA system model [19]. A frame consists of N slots of equal length, and M users each have one packet to send in the frame. The offered load is

$$G = \frac{M}{N} \quad \text{(users per slot)}. \tag{4.1}$$

Each user independently draws a repetition degree ℓ from the distribution $\Lambda(x)$ and transmits ℓ replicas of its packet in ℓ distinct slots chosen uniformly at random. The frame is represented by a binary user-by-slot matrix $\mathbf{M}$ whose entry $m_{u,s}$ equals 1 when user u has a replica in slot s. The receiver observes, for each slot, the set of replicas that occupy it. A slot with exactly one replica (a *singleton*) yields a decoded packet; a slot with two or more replicas is a collision. Once a user is decoded, all its replicas are cancelled, which reduces the occupancy of every slot that it used. The procedure is repeated until no singleton remains or all users are decoded.

Three reference schemes are evaluated within the same framework so that the comparison is made under identical conditions:

- **Frame Slotted ALOHA:** every user sends a single packet, $\Lambda(x) = x$. Cancellation then has no effect, because a decoded user occupies no other slot.
- **CRDSA:** every user sends exactly two replicas, $\Lambda(x) = x^2$, with SIC.
- **IRSA without SIC:** the IRSA replicas are placed as in the main scheme, but only users that have at least one singleton replica in the original frame are counted as decoded. This isolates the contribution of SIC.

4.6 Software Tools

KiCad is used for schematic capture, footprint assignment, PCB routing and 3D visualisation of the tag [24].

CST Studio Suite is used for full-wave electromagnetic simulation of the tag under plane-wave illumination and for the computation of RCS versus frequency for the two load conditions [25].

GNU Radio Companion is used to build the three-tag signal model from signal sources, vector sources, filters, arithmetic blocks, a noise source, synchronisation and slicing blocks, a BER block and Qt GUI sinks for time-domain, frequency-domain and numerical outputs [21].

Python 3 with **NumPy** is used for the standalone IRSA/SIC implementation [22]. NumPy's random generator (`default_rng`) provides reproducible random numbers, and integer arrays hold the user-by-slot matrix so that slot occupancies are obtained as column sums. **Matplotlib** is used to plot throughput and PLR against offered load [23]. Python suits this task because the simulation needs only array operations and random sampling, which NumPy provides directly, and because GNU Radio's embedded blocks are also written in Python, which eases later integration.

4.7 Simulation Parameters and Assumptions

4.7.1 CST Electromagnetic Simulation

Table 4.3 – CST simulation parameters

Parameter	Value
Structure	Dipole tag imported from the KiCad design
Frequency sweep	120–280 MHz
Excitation	X-polarised plane wave
Electric-field amplitude	1 V/m
ON-state load	2 Ω resistive
OFF-state load	0.45 pF capacitive
Evaluated quantity	RCS versus frequency

4.7.2 GNU Radio Signal Model

Table 4.4 – GNU Radio model parameters

Parameter	**Value**
Number of tags	3
Sampling rate, F_s	50 kS/s
Symbol rate, R_s	20 symbols/s
Samples per symbol, L	2500
Carrier (each tag)	Cosine, 10 kHz, amplitude 1
Subcarrier control (Tags 1, 2, 3)	Square wave, 1 kHz, 800 Hz, 1.2 kHz
Data sequence length (Tags 1, 2, 3)	8, 7 and 11 bits (repeating)
Tag response coefficients	$\sigma_{ON} = 1.385 \times 10^{-3}$, $\sigma_{OFF} = 174 \times 10^{-6}$
Scale factor, carrier path	0.0414
Scale factor, tag outputs	0.01166
Noise source	Gaussian, amplitude setting 27.9×10^{-6}

The response coefficients and scale factors are numerical settings of the model. They set the relative levels of the carrier, the tag responses and the noise inside the flowgraph and are kept separate from the physical RCS values of the CST study.

4.7.3 Standalone IRSA/SIC Simulation

Table 4.5 lists the parameters of the packet-level simulation, taken directly from the project code (Appendix C).

Table 4.5 – Parameters of the standalone IRSA/SIC simulation

Parameter	**Value**
Number of slots per frame, N	200
Offered load, G	0.10 to 1.00 in steps of 0.05 (19 load points)
Number of users, M	$\max\{1, \text{round}(G \cdot N)\}$, i.e. 20 to 200
Repetition-degree distribution	$\Lambda(x) = 0.5x^2 + 0.28x^3 + 0.22x^8$
Average replicas per user, $\bar{\ell}$	$2(0.5) + 3(0.28) + 8(0.22) = 3.6$
Slot selection	ℓ distinct slots, uniform without replacement
Monte Carlo trials per load	800 independent frames
Random seeds	Master seed 0; one derived seed per frame
Channel model	Collision channel: singleton decoded, collision lost
Receiver	Iterative SIC with ideal cancellation
Baselines	FSA ($\Lambda(x) = x$), CRDSA ($\Lambda(x) = x^2$), IRSA without SIC
Metrics	Mean throughput T and mean packet loss rate (PLR)

The simulation rests on the following assumptions:

1. **Slot synchronisation.** All users are aligned to the slot boundaries of the frame, and every replica occupies exactly one slot.
2. **One packet per user per frame.** Each of the M users is active and has exactly one packet; there is no retransmission across frames.
3. **Collision channel without capture.** A slot holding one replica is always decoded, and a slot holding two or more replicas is never decoded directly, regardless of power differences. Noise-induced errors in clean slots are not modelled.
4. **Known replica positions.** When a packet is decoded, the receiver knows the slots of its other replicas, as when each replica carries pointers to its twins [18].
5. **Ideal cancellation.** The contribution of a decoded replica is removed completely from its slot, leaving no residual interference.

4.8 Separation of the Simulation Methodologies

The GNU Radio model and the Python simulation work at different levels of abstraction and answer different questions. Packet-level collision resolution was not performed inside GNU Radio; the two simulations are independent. Table 4.6 sets out the differences.

Table 4.6 – Comparison of the signal-level and packet-level simulation methodologies

Aspect	GNU Radio signal model	Python IRSA/SIC simulation
Purpose	Recover the bits of three tags from a composite noisy waveform	Measure how many packets per slot a random-access scheme delivers
Unit of data	Signal samples at 50 kS/s	Packet replicas in slots of a frame
Multi-tag method	Fixed subcarrier per tag (FDMA)	Random slot selection with repetition (IRSA)
Channel	Additive Gaussian noise on a summed waveform	Collision channel with ideal SIC
Tags or users	3	20 to 200 per frame
Outputs	Spectra, waveforms, symbols, SNR, BER	Throughput, PLR, peak throughput
Tool	GNU Radio Companion	Python with NumPy and Matplotlib

The two models are complementary: the signal model demonstrates that a receiver can extract the bits of individual tags from the summed reflection, which a "decoded singleton slot" assumes at packet level, and the packet model quantifies how many tags can be served per slot once such decoding is available.

4.9 Performance Metrics

RCS contrast. The difference $\Delta\sigma = \sigma_{ON} - \sigma_{OFF}$ (dB) between the ON-state and OFF-state RCS, both in dBsm, at a specified frequency.

Bit error rate. The ratio $\mathrm{BER} = N_{\mathrm{err}}/N_{\mathrm{bits}}$ of mismatched bits to compared bits.

Throughput. For a single frame with D decoded users,

$$T = \frac{D}{N} \quad \text{(decoded packets per slot)}, \tag{4.2}$$

averaged over all trials at each load.

Packet loss rate. For a single frame,

$$\text{PLR} = \frac{M - D}{M}, \tag{4.3}$$

also averaged over the trials. From Equations (4.1), (4.2) and (4.3), the two packet-level metrics are related by $T = G(1 - \text{PLR})$ for each frame.

CHAPTER 5

IMPLEMENTATION

5.1 Implementation Environment and Setup

The implementation follows the four workstreams. The tag was implemented as a KiCad schematic and PCB layout and then imported into CST Studio Suite for electromagnetic simulation. The three-tag signal model was implemented as a GNU Radio Companion flowgraph. The packet-level access scheme was implemented as a standalone Python program using NumPy and Matplotlib. Each implementation is described in turn below, followed by the verification procedures, the challenges met during the work and the implementation status.

5.2 KiCad Hardware Implementation

The tag schematic (Figure 5.1) contains the four-pin connector J1, the TPS7A20 regulator (U2), the SN74LVC1T45 level translator (U3), the ADG902 RF switch (U1), decoupling capacitors and the antenna connection pads U4 and U5. The control signal from the external ESP32-S3 enters through J1 pin 3 and reaches translator input A. The DIR input is tied high to fix the A-to-B direction, and the translated output B drives the CTRL pin of the ADG902. The switch is powered from the regulated 2.5 V rail and shares a common ground with the controller. The ADG902 does not generate any signal of its own; it changes the termination seen by the antenna, so that the tag modulates the reflection of an incident wave.

5.2.1 Power and Control Pathways

The 5 V input at J1 pin 1 feeds the TPS7A20, which provides a nominal 2.5 V output, with 1 μF capacitors at its input and output. J1 pin 2 supplies 3.3 V to the input-side logic domain of the SN74LVC1T45, so the translator input matches the ESP32-S3 GPIO level. The translator output drives the ADG902 CTRL pin at 2.5 V, and a 100 nF decoupling capacitor sits close to the switch. J1 pin 4 is the common reference for the supplies and the control signal. Because the controller is on its own board, the tag PCB stays small and contains only the parts needed for the RF switching function.

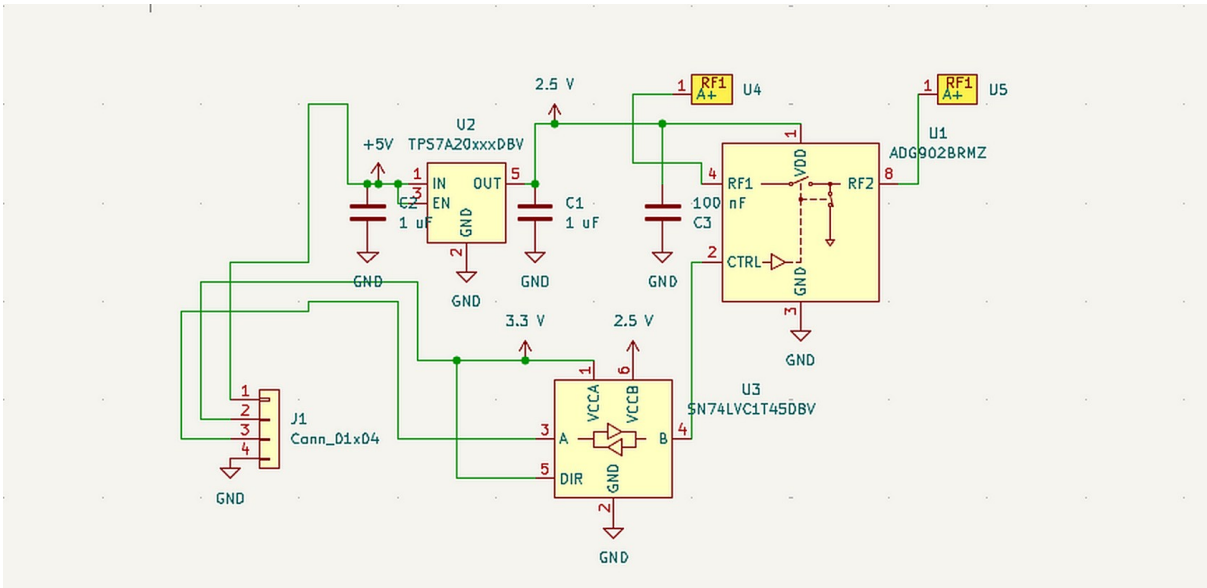


Figure 5.1 – Schematic of the RF backscatter tag with the regulator, level translator, RF switch and connector interface

5.2.2 PCB Layout and 3D Model

The routed layout and its 3D view (Figure 4.1) place the RF switch next to the antenna pads to keep the RF path short, with the regulator, translator and passive components grouped behind it and the connector J1 at the board edge. The 3D model was used to check component placement, connector orientation and clearances before the design was exported for electromagnetic analysis.

5.3 CST Electromagnetic Simulation

The tag design was imported into CST Studio Suite, and the dipole structure was simulated over 120–280 MHz (Figure 5.2). The excitation was an X-polarised plane wave with an electric-field amplitude of 1 V/m. The switch was represented by a lumped element at the feed, set to 2 Ω for the ON state and 0.45 pF for the OFF state, and a separate run was carried out for each state. In each run the RCS was recorded as a function of frequency.

The two RCS curves are plotted together in Figure 5.3. The upper curve corresponds to the ON state and the lower curve to the OFF state. The ON-state RCS reaches its maximum of about $+1.5$ dBsm at 220 MHz, while the OFF-state RCS at the same frequency is about -16.7 dBsm. Over the whole sweep the ON state scatters more strongly than the OFF state, and the gap is largest near 220 MHz. The OFF-state curve rises towards higher frequencies and reaches about -10.7 dBsm at 280 MHz. The results at 220 MHz are analysed in Chapter 6.

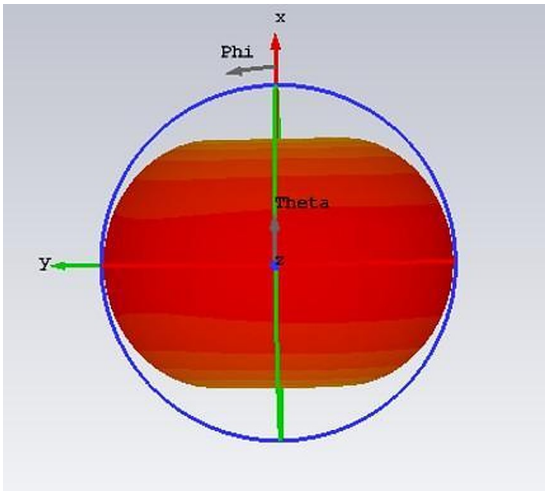


Figure 5.2 – CST simulation set-up and model configuration of the dipole tag

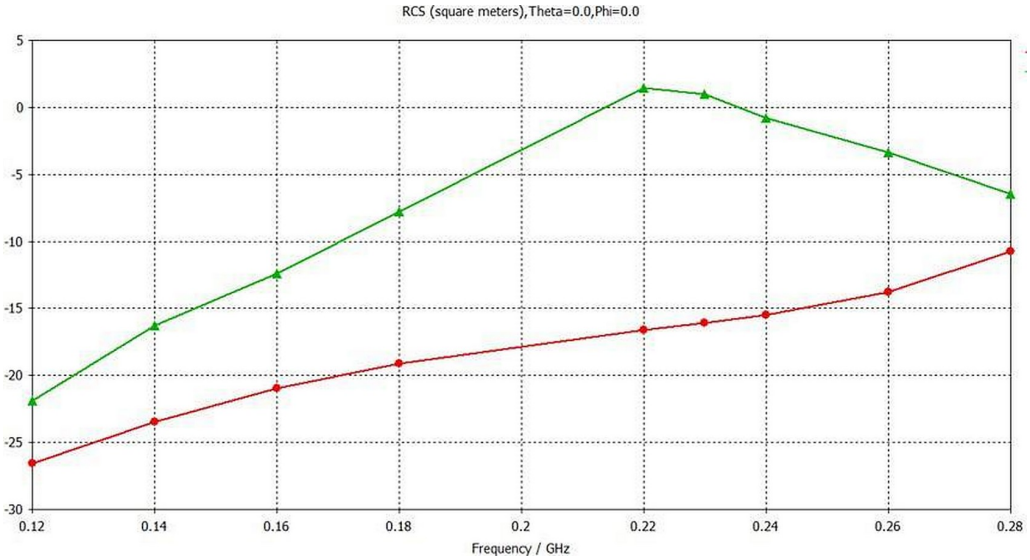


Figure 5.3 – CST-derived RCS versus frequency for the ON-state (2 Ω) and OFF-state (0.45 pF) load conditions

5.4 GNU Radio Signal Model

The GRC flowgraph models three tag branches (Figure 5.4). Each tag has a carrier source and a square-wave subcarrier control. The three carriers are 10 kHz cosines of amplitude 1 at a sampling rate of 50 kS/s, and the control sources run at 1 kHz, 800 Hz and 1.2 kHz (Figure 5.5). The high level of the square wave represents the ON state and the low level the OFF state, and the control selects the corresponding coefficient in the Backscatter Tag block. Because each control frequency is added to the 10 kHz carrier, the three tags occupy distinct spectral regions around 11 kHz, 10.8 kHz and 11.2 kHz, which is the subcarrier separation of Section 2.3.2.

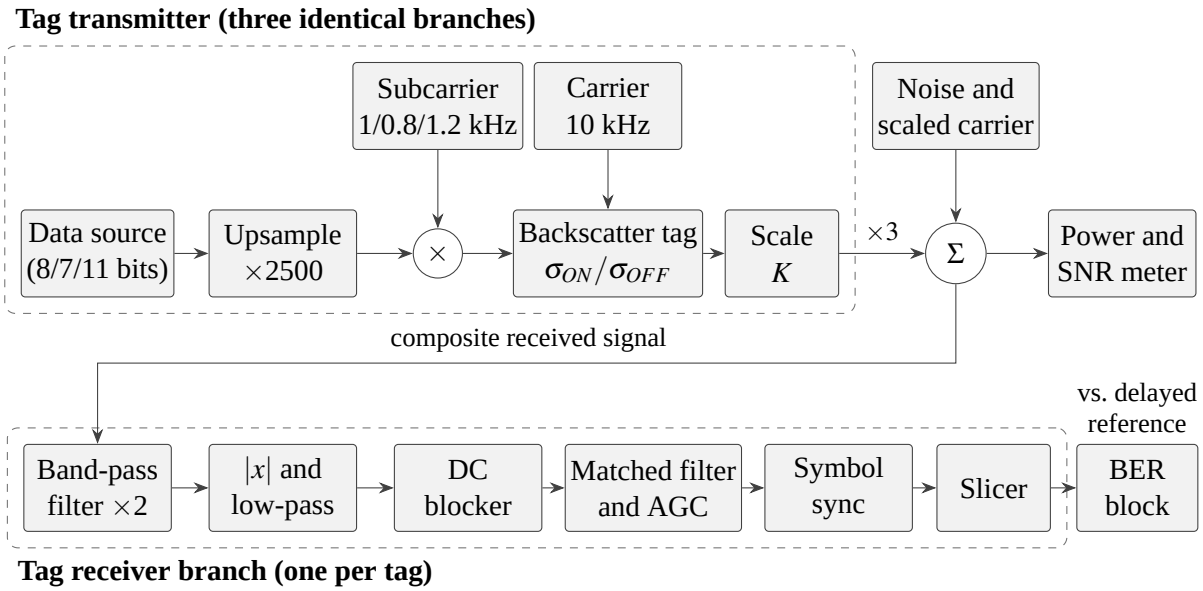


Figure 5.4 – Overall structure of the three-tag GNU Radio signal model (simplified block diagram)

5.4.1 Baseband Data Generation

A Vector Source block generates a repeating binary sequence for each tag, of lengths 8, 7 and 11 bits for Tags 1, 2 and 3. A Char-to-Float block converts the byte values to floating-point samples, and an Interpolating FIR filter with unit taps raises the sample rate by a factor of 2500, holding each bit for one symbol period. A Virtual Sink then passes the baseband data to the matching Virtual Source in the tag path (Figure 5.6). With $F_s = 50\,000$ samples/s and $R_s = 20$ symbols/s, the number of samples per symbol is

$$L = \frac{F_s}{R_s} = 2500. \tag{5.1}$$

5.4.2 Signal Multiplication, Scaling and Combining

In each branch the square-wave control is multiplied by the baseband data before it enters the Backscatter Tag block:

$$c[n] = d[n] \cdot q[n], \tag{5.2}$$

where $d[n]$ is the data and $q[n]$ the subcarrier control. The resulting signal switches at the subcarrier rate during a “1” and stays in the OFF state during a “0”, which is OOK of the subcarrier. The Backscatter Tag block selects σ_{ON} when its control input exceeds 0.5 and σ_{OFF} otherwise, and multiplies the selected coefficient by the carrier:

$$y[n] = \sigma[n] \cdot x_c[n]. \tag{5.3}$$

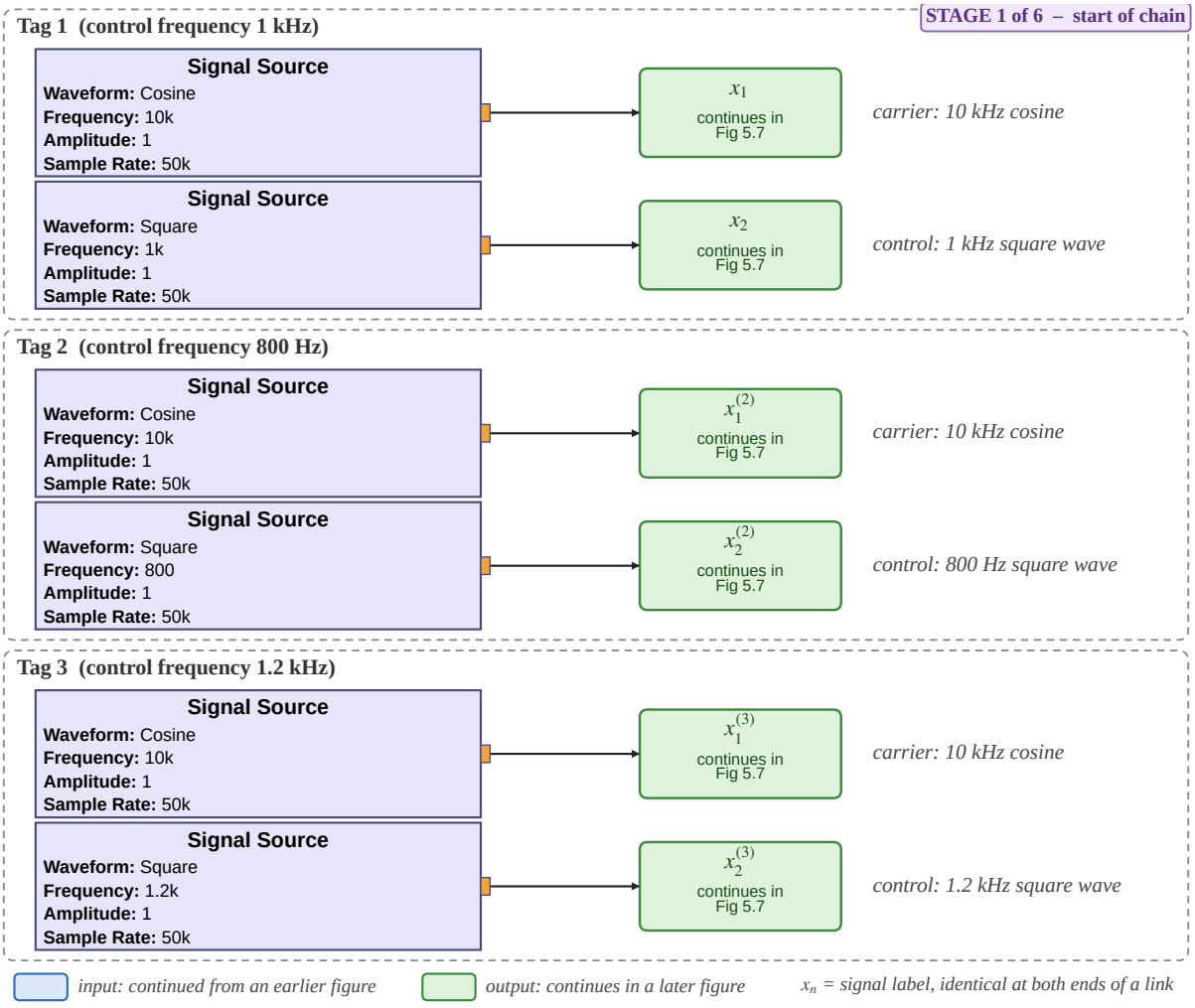


Figure 5.5 – Carrier and control-signal generation for the three simulated tags

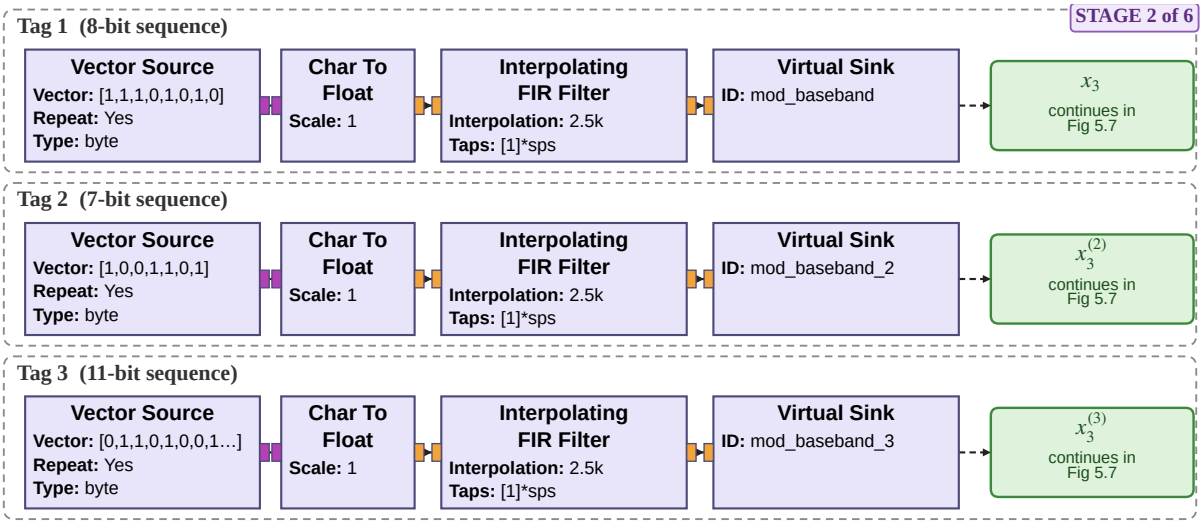


Figure 5.6 – Baseband data generation using Vector Source, Char-to-Float and Interpolating FIR filter blocks

Multiply Const blocks scale the signals as $y[n] = K\,x[n]$, with $K = 0.0414$ for the carrier path and $K = 0.01166$ for the tag outputs. A Gaussian noise source with an amplitude setting of 27.9×10^{-6} is added, and the Add block sums the carrier term, the three scaled tag responses and the noise sample by sample to form one composite received signal (Figure 5.7).

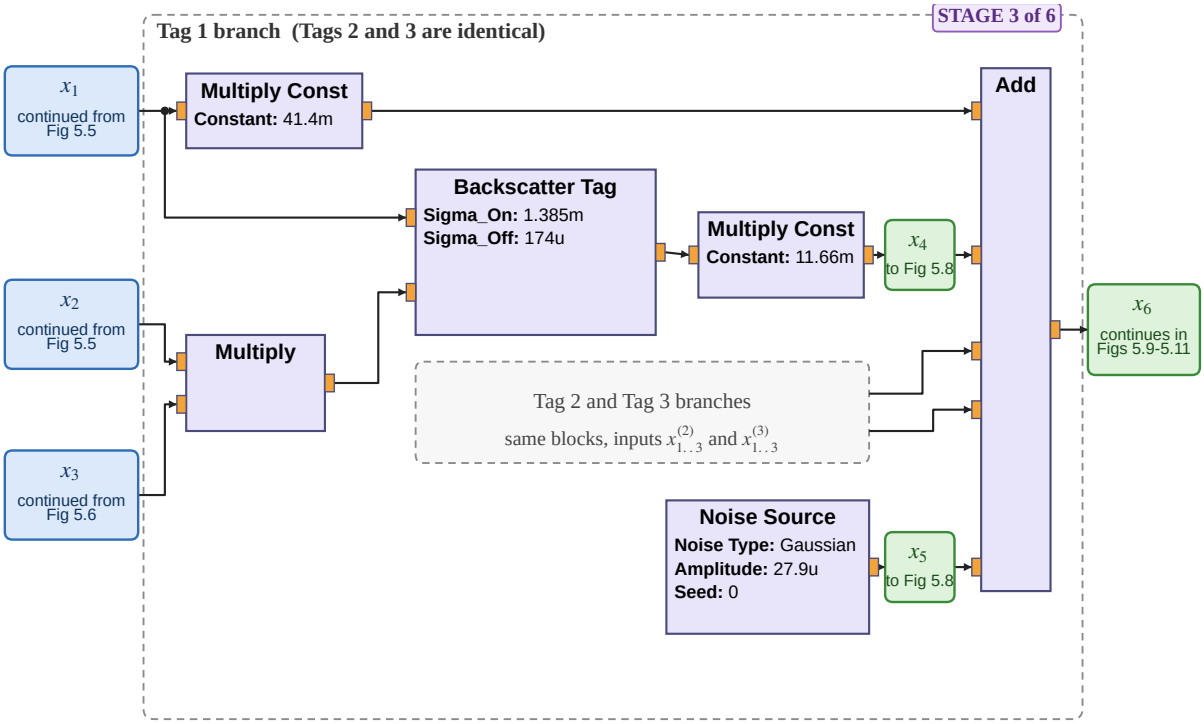


Figure 5.7 – Signal multiplication, tag-response modelling, scaling, noise addition and signal combining

5.4.3 Power and SNR Measurement

Two parallel branches estimate the signal and noise powers (Figure 5.8). In each branch a Multiply block squares its input and a Moving Average block averages the squared samples:

$$P = \frac{1}{N_a} \sum_{n=0}^{N_a-1} x^2[n]. \tag{5.4}$$

The upper branch measures the backscatter signal power and the lower branch the noise power. A Divide block forms the linear SNR, and a Log10 block converts it to decibels:

$$\mathrm{SNR} = \frac{P_{\mathrm{signal}}}{P_{\mathrm{noise}}}, \qquad \mathrm{SNR}_{\mathrm{dB}} = 10\log_{10}(\mathrm{SNR}). \tag{5.5}$$

Qt GUI Number Sinks report the backscatter power, the noise power and the instantaneous SNR during the run.

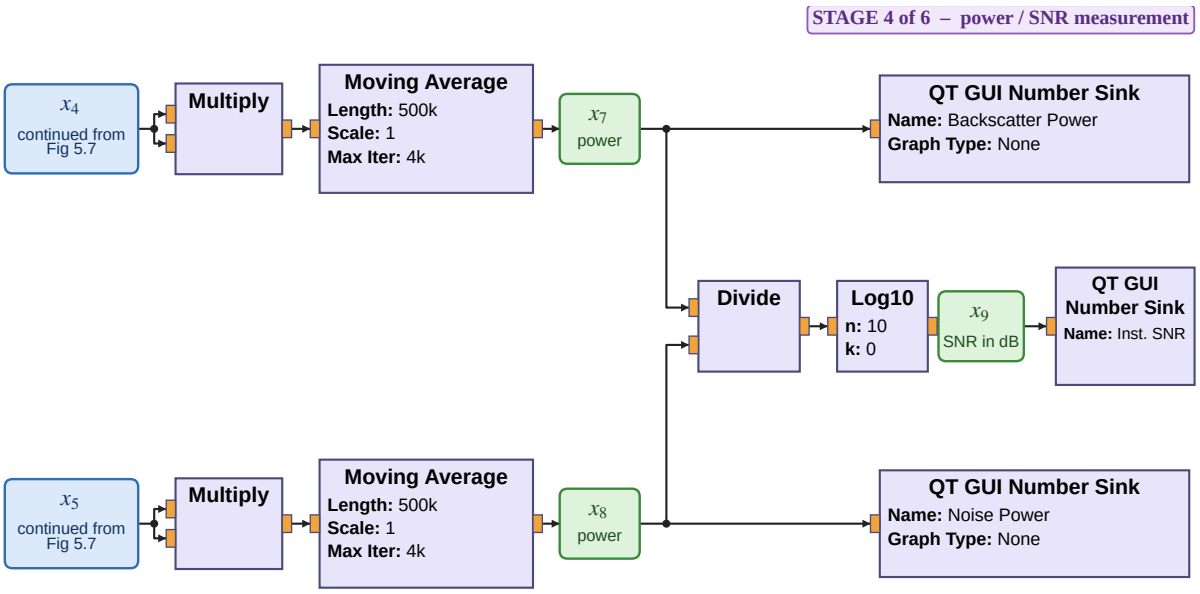


Figure 5.8 – Combined-signal output and power/SNR measurement branches in the flowgraph

5.4.4 Receiver Processing

The receiver separates the composite signal into tag-specific paths (Figure 5.9). For Tag 1, two cascaded band-pass filters select the band around 11 kHz with cut-off frequencies of 10.92 kHz and 11.08 kHz, giving

$$f_c = \frac{f_H + f_L}{2} = 11 \text{ kHz}, \qquad BW = f_H - f_L = 160 \text{ Hz}. \tag{5.6}$$

Cascading the two filters gives a steeper roll-off and stronger rejection of the neighbouring subcarriers and of the carrier. The subsequent stages are:

1. **Absolute value:** rectifies each sample and acts as an envelope (energy) detector.
2. **Low-pass filter:** 40 Hz cut-off; smooths the rectified signal.
3. **DC blocker:** length 5000; removes the constant offset. The length was tuned using the filter response.
4. **Decimating FIR filter:** 2500 taps of value $1/2500 = 0.0004$; a moving-average matched filter for rectangular symbols.
5. **AGC:** initial gain 1 and maximum gain 65 536; normalises the level before timing recovery.

6. **Symbol synchronisation:** 2500 samples per symbol, aligning the sampling instants with the symbol timing so that $R_s = F_s/\text{sps} = 20$ symbols/s.
7. **Binary slicer:** converts the synchronised samples into bit decisions and sends them to a Virtual Sink named `bits_out`.
8. **Monitoring:** a Qt GUI Frequency Sink records the spectrum after the second band-pass filter, and Qt GUI Time Sinks record the signals at the receiver stages.

The Tag 1 chain is therefore: combined signal → band-pass filters → Abs and low-pass filter → DC blocker → matched FIR filter → AGC → symbol synchronisation → binary slicer → recovered bits. This sequence closely follows the receiver of the base thesis, which also uses band-pass filtering at the subcarrier, magnitude detection, a one-symbol matched filter and an AGC with threshold decision [1].

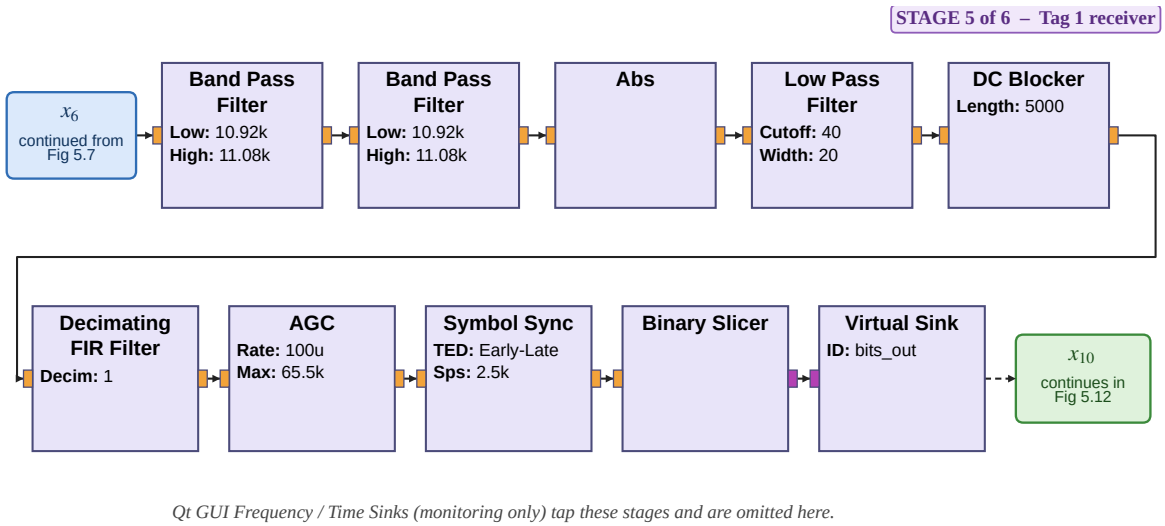


Figure 5.9 – Tag 1 receiver-processing flowgraph

The Tag 2 and Tag 3 receivers use the same stages and differ only in their band-pass settings, which follow each tag's subcarrier frequency (Table 5.1, Figures 5.10 and 5.11).

Table 5.1 – Band-pass filter settings of the tag receiver branches

Tag	Control freq.	Centre freq.	Lower cut-off	Upper cut-off	Bandwidth
1	1 kHz	11 kHz	10.92 kHz	11.08 kHz	160 Hz
2	800 Hz	10.8 kHz	10.72 kHz	10.88 kHz	160 Hz
3	1.2 kHz	11.2 kHz	11.12 kHz	11.28 kHz	160 Hz

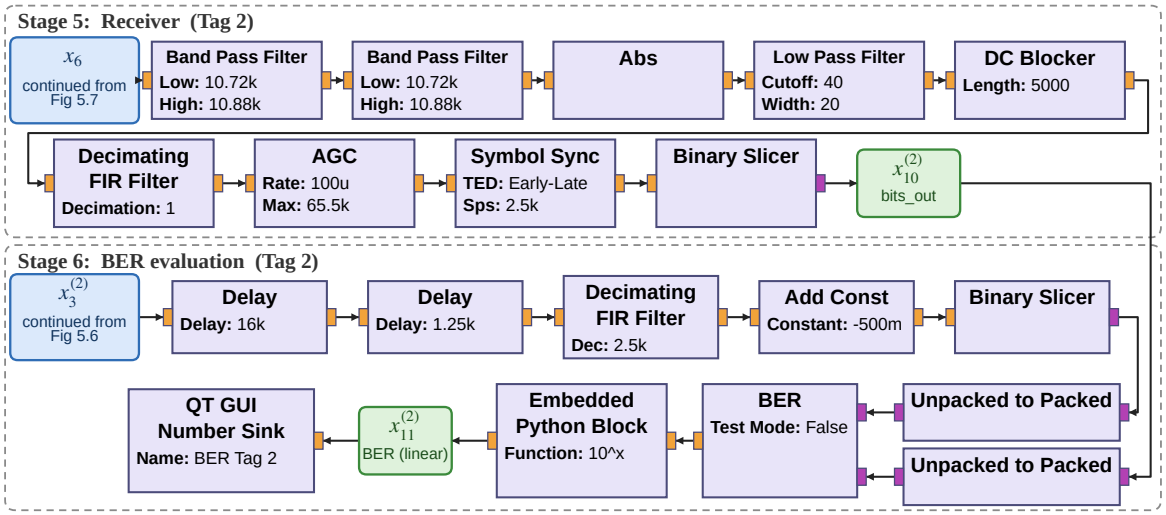


Figure 5.10 – Tag 2 receiver-processing and BER-evaluation flowgraph

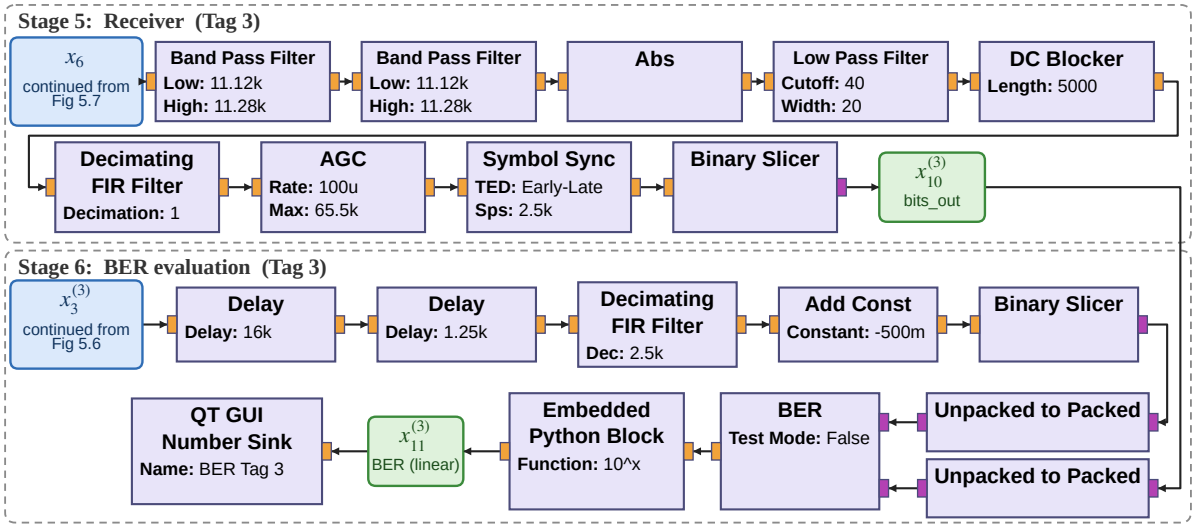


Figure 5.11 – Tag 3 receiver-processing and BER-evaluation flowgraph

5.4.5 BER Evaluation

The BER branch (Figure 5.12) compares the recovered bits with a reference derived from the original baseband data:

1. **Reference generation:** the original data are taken from the Virtual Source `mod_baseband`. A Delay block of 16 000 samples aligns the reference with the recovered bits, and a second Delay block of 1250 samples moves the sampling point to the middle of each bit, since 1250 is half of the 2500 samples per symbol.
2. **Reference sampling:** a Decimating FIR filter with a decimation factor of 2500 reduces the reference to one sample per symbol, giving $F_{out} = 50\,000/2500 = 20$ samples/s.
3. **Add Const and Binary Slicer:** adding -0.5 maps the reference values from $\{0, 1\}$ to $\{-0.5, +0.5\}$, and the slicer converts them back to bit decisions.

4. **Unpacked-to-Packed blocks:** one block receives the reference bits and the other receives the recovered bits from the Virtual Source `bits_out`, preparing both streams for comparison.
5. **BER block:** compares the reference and recovered bits, counts the mismatches and outputs the error rate in logarithmic form.
6. **Embedded Python blocks:** one block applies 10^x to the BER block output to convert it to a linear BER; another generates the BER-versus-SNR data for different noise settings.
7. **Qt GUI Number Sink:** reports the resulting BER of each tag.

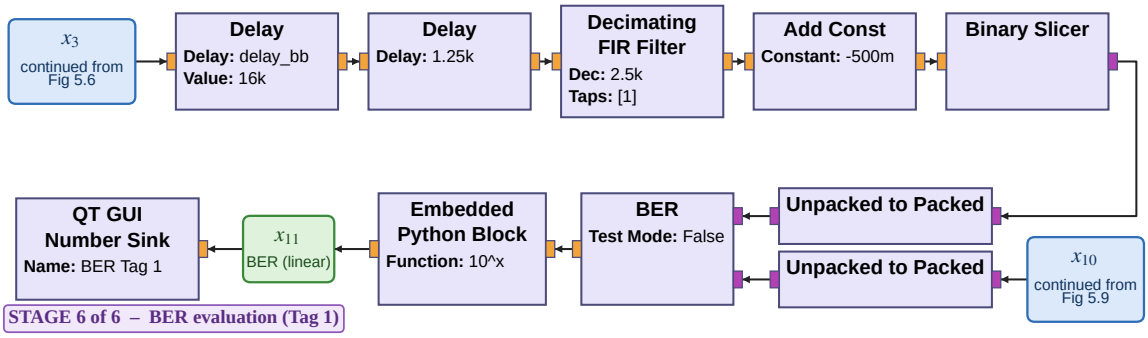


Figure 5.12 – Tag 1 BER-evaluation flowgraph with reference generation, delay alignment and the BER block

5.5 Standalone Implementation of IRSA and SIC

IRSA and SIC were implemented independently in Python to investigate packet-level multiple access and collision resolution for a large population of tags. The program is self-contained: it generates frames, decodes them, averages the results over many frames and sweeps the offered load. It does not use the RF tag hardware or the GNU Radio signal model, and its inputs and outputs are not exchanged with them. Its role is to quantify the gain that IRSA with SIC offers over plain random access under the standard packet-level assumptions of Section 4.7.3. The complete source code is listed in Appendix C.

5.5.1 Software Structure

Table 5.2 lists the functions of the program and their roles. Figure 5.13 outlines the overall flow.

Table 5.2 – Functions of the standalone IRSA/SIC program

Function	Role
`build_transmit_matrix`	Draws a repetition degree for each user and places its replicas in distinct random slots; returns the user-by-slot matrix **M**
`irsa_sic_decode`	Runs iterative SIC on a copy of **M** and returns the list of decoded users
`simulate_frame`	Builds and decodes one frame; returns its PLR and throughput
`monte_carlo`	Repeats `simulate_frame` for a specified number of trials with derived seeds; returns the mean PLR and throughput at one load
Main script	Sweeps the load, collects the results and plots PLR and throughput against load

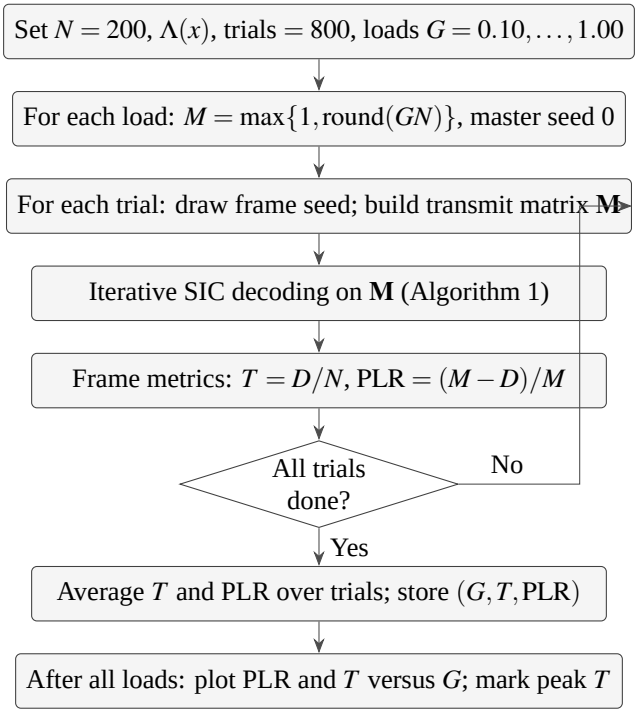


Figure 5.13 – Flow of the standalone IRSA/SIC simulation

5.5.2 IRSA Packet Repetition and Slot Allocation

Each user in a frame represents one tag with one packet. The function `build_transmit_matrix` first converts the degree distribution into two arrays, the degrees $\{2,3,8\}$ and their probabilities $\{0.5,0.28,0.22\}$, and normalises the probabilities to sum to one. For each user u it then:

1. draws a degree d_u from $\{2,3,8\}$ with the probabilities above, using `rng.choice`;
2. limits the degree to the frame size, $d_u \leftarrow \min(d_u, N)$;
3. selects d_u distinct slot indices uniformly at random from $\{0,\ldots,N-1\}$, using `rng.choice` without replacement; and
4. sets $m_{u,s} = 1$ for each selected slot s.

The matrix $\mathbf{M}$ therefore has one row per user and one column per slot, and row u contains exactly d_u ones. With the distribution of Table 4.5, half of the users send two replicas, 28% send three and 22% send eight, so a user sends 3.6 replicas on average. At load G the frame carries on average $3.6G$ replicas per slot; for example, about 2.9 replicas per slot at $G = 0.8$. The high-degree users are what drive the decoding: their many replicas make it likely that at least one lands in a clean slot early in the process, and cancelling them clears several slots at once.

5.5.3 Collision Modelling and SIC Decoding

Collisions are represented by the column sums of $\mathbf{M}$. The occupancy of slot s is

$$c_s = \sum_u m_{u,s}, \tag{5.7}$$

so $c_s = 0$ is an idle slot, $c_s = 1$ a singleton and $c_s \geq 2$ a collision. The function `irsa_sic_decode` works on a copy of $\mathbf{M}$ and repeats the steps in Algorithm 1.

Three details of the implementation matter. First, all singletons found in one pass are decoded together, and the cancellations are applied together before the occupancies are recomputed, so one iteration of the loop corresponds to one round of parallel decoding. Second, a user that appears as a singleton in two different slots is added to `new_users` only once, because `new_users` is a set. Third, cancellation is modelled by setting the user's whole row to zero, which removes its replicas from every slot at once; this is the ideal-cancellation assumption. The loop ends when no singleton is left, which leaves the remaining users undecoded, or when every user has been decoded.

Algorithm 1: Iterative SIC decoding as implemented in `irsa_sic_decode`

Input: Transmit matrix M with M users (rows) and N slots (columns)
Output: Sorted list of decoded users

```
decoded = empty list;
while True do
    col_sums = sum of each column of M (number of replicas in each slot);
    singletons = list of slots s where col_sums[s] == 1;
    if singletons is empty then stop;
    new_users = empty set;
    foreach slot s in singletons do
        u = the only user with M[u, s] == 1;
        add u to new_users;
    end
    if new_users is empty then stop;
    foreach user u in new_users do
        append u to decoded;
        set every entry in row u of M to 0 (cancel all replicas of u);
    end
    if length of decoded == M then stop;
end
return sorted(decoded);
```

Table 5.3 traces this procedure for a small frame of four users and five slots, with user 1 in slots 1 and 3, user 2 in slots 3 and 4, user 3 in slots 2, 4 and 5, and user 4 in slots 2 and 5. The corresponding slot occupancy is drawn in Figure 5.14. Initially only slot 1 is a singleton, so a receiver without SIC would decode only user 1. With SIC, each cancellation exposes a new singleton, and all four users are recovered in four iterations.

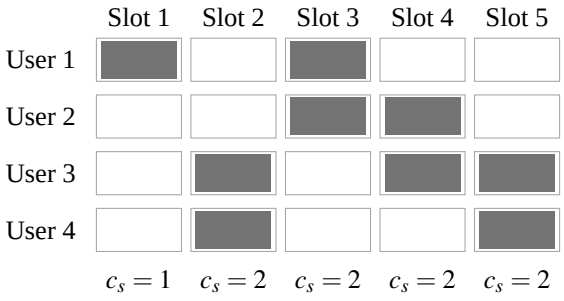


Figure 5.14 – Example frame for SIC decoding: shaded cells are packet replicas, c_s is the initial slot occupancy

The example also illustrates how decoding can fail. If users 3 and 4 had both chosen exactly slots 2 and 5, neither would ever appear alone in a slot, and SIC could not separate them at

Table 5.3 – SIC iterations for the example frame of Figure 5.14

Iteration	**Singleton slots**	**Decoded**	**Occupancy after cancellation** $(c_1, \ldots, c_5)$
0	–	–	(1, 2, 2, 2, 2)
1	1	User 1	(0, 2, 1, 2, 2)
2	3	User 2	(0, 2, 0, 1, 2)
3	4	User 3	(0, 1, 0, 0, 1)
4	2, 5	User 4	(0, 0, 0, 0, 0)

any stage. Such configurations are the stopping sets discussed in Section 2.13, and they are the source of the residual packet loss at low load reported in Chapter 6.

5.5.4 Throughput Calculation

The function `simulate_frame` creates a random generator from the frame seed, builds **M**, decodes it, and computes the frame metrics from the number of decoded users D:

$$T_{\text{frame}} = \frac{D}{N}, \qquad \text{PLR}_{\text{frame}} = \frac{M - D}{M}. \tag{5.8}$$

A packet counts as successfully decoded when its user index is in the list returned by `irsa_sic_decode`, that is, when at least one of its replicas became a singleton at some stage of the iterative process. Each user is counted once, regardless of how many of its replicas were cancelled. The throughput is expressed in decoded packets per slot. The repetitions are not counted as additional throughput, so the ideal limit is $T = G$ and the absolute maximum is one packet per slot.

The function `monte_carlo` creates a master generator with seed 0, draws one integer seed per trial from it, calls `simulate_frame` with that seed, and returns the means

$$\bar{T}(G) = \frac{1}{K} \sum_{k=1}^{K} T_{\text{frame}}^{(k)}, \qquad \overline{\text{PLR}}(G) = \frac{1}{K} \sum_{k=1}^{K} \text{PLR}_{\text{frame}}^{(k)}, \tag{5.9}$$

with $K = 800$ trials. The main script repeats this for the 19 loads from 0.10 to 1.00, with $N = 200$ slots, and records $(G, \bar{T}, \overline{\text{PLR}})$ for each. Because the master seed is the same at every load, the results are fully reproducible.

5.5.5 Baseline Schemes under Identical Conditions

The baselines were obtained with the same functions and settings by changing only the degree distribution or the decoding rule. Frame Slotted ALOHA uses `degree_dist = {1: 1.0}`, and CRDSA uses `degree_dist = {2: 1.0}`. For IRSA without SIC, the transmit matrix is built exactly as for IRSA, and a user is counted as decoded when it has at least one replica in a slot whose initial occupancy is one; no cancellation is performed. All baselines use $N = 200$, the same 19 loads, 800 trials per load and master seed 0.

5.6 Implementation Status

The KiCad design is complete, with schematic, routed layout and 3D model. The CST simulation of both load states is complete. The GNU Radio model with three tags, separate receivers, BER evaluation and a BER-versus-SNR sweep is complete. The standalone packet-level simulation of IRSA with SIC is complete, including its throughput and PLR evaluation and the three baselines. Fabrication of the tag, the over-the-air link, and the integration of the IRSA/SIC procedure with the GNU Radio model and the physical backscatter system remain pending, as defined by the system boundary in Table 3.3.

5.7 Testing and Verification Methodology

Each workstream was checked with tests suited to its level:

- **Hardware design:** the schematic connections were reviewed against the pin functions of the TPS7A20, SN74LVC1T45 and ADG902, and the 3D model was used to check placement and connector orientation.
- **CST model:** the two load states were simulated under the same excitation and frequency sweep, so that the difference between the curves reflects only the change of load.
- **GNU Radio model:** the spectrum after each band-pass filter was checked for the expected subcarrier lines, the time-domain traces were used to tune the 16 000-sample alignment delay before any BER was recorded, and the noise amplitude was varied to confirm that the BER responds to the SNR.
- **IRSA/SIC simulation:** the FSA baseline was compared with the closed-form result of Equation (2.6). For $N = 200$ the formula gives 0.3044 at $G = 0.5$ and 0.3688 at $G = 1.0$, and the simulation gives 0.3035 and 0.3678, a difference of about 0.3%. This agreement confirms the slot-selection and singleton-detection logic. In addition, at low load the

IRSA throughput equals the offered load to three decimal places, as expected when almost every user is recovered.

5.8 Implementation Challenges and Solutions

Logic-level mismatch. The ESP32-S3 GPIO operates at 3.3 V, while the ADG902 control input is referenced to its 2.5 V supply. The SN74LVC1T45 dual-supply translator, with its A side at 3.3 V and its B side at 2.5 V, resolves this mismatch.

Subcarrier separation. The three subcarriers lie only 200 Hz apart. Two cascaded band-pass filters with a 160 Hz bandwidth were used to obtain sufficient rejection of the neighbouring tags.

Receiver delay alignment. The filters, DC blocker and symbol synchroniser introduce a delay between the transmitted and recovered bits. The reference path was delayed by 16 000 samples, tuned with the time-domain traces, and by a further 1250 samples to sample each reference bit at its centre.

DC offset. The strong carrier term produces a large DC level after envelope detection. A DC blocker of length 5000 removes it while preserving the 20 symbol/s data.

Monte Carlo run time and reproducibility. The full sweep requires $19 \times 800 = 15\,200$ frames per scheme. Vectorised column sums keep each SIC iteration inexpensive, and seeding each frame from a master seed makes every run repeatable.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Experimental Setup and Test Conditions

The results in this chapter come from the four workstreams under the conditions set out in Chapter 4. The hardware result is the completed tag design. The electromagnetic results are the CST RCS curves for the ON (2 Ω) and OFF (0.45 pF) loads under 1 V/m plane-wave illumination over 120–280 MHz. The signal-level results come from the three-tag GNU Radio model with the parameters of Table 4.4, first in a baseline run and then in a sweep of the noise amplitude. The packet-level results come from the Python simulation with $N = 200$ slots, 19 loads between 0.10 and 1.00 and 800 frames per load, for IRSA with SIC and for the three baselines.

6.2 PCB Design Outcome

The KiCad design defines a compact tag board that carries only the RF switching function and its supporting circuitry: the ADG902 switch next to the antenna pads, the TPS7A20 regulator providing the 2.5 V rail, the SN74LVC1T45 translator adapting the 3.3 V control logic, and decoupling capacitors (Figures 4.1 and 5.1). The external ESP32-S3 controller connects through J1, which carries 5 V, 3.3 V, the control line and ground. Keeping the controller off the tag board separates the digital and RF sections and lets the same tag be driven by any controller with a 3.3 V output. The design is ready for fabrication.

6.3 CST RCS Results

The CST results (Figure 5.3) give a frequency-dependent scattering response for both load conditions over the 120–280 MHz sweep, with the ON state scattering more strongly at every frequency. The values at 220 MHz, where the separation is largest, are listed in Table 6.1.

An RCS contrast of 18.2 dB corresponds to a ratio of about 66 between the scattered powers of the two states. By the radar equation (2.4), for a fixed emitter–tag–reader geometry the backscattered power at the reader changes by the same 18.2 dB when the tag switches state. The tag therefore produces a deep amplitude modulation of its reflection, which is the property an

Table 6.1 – Simulated RCS at 220 MHz for the two load conditions

State	Load	RCS (dBsm)
ON	2 Ω resistive	$\approx +1.5$
OFF	0.45 pF capacitive	≈ -16.7
Difference (ON − OFF)		≈ 18.2 dB

OOK backscatter link relies on. The ON-state curve peaks near 220 MHz, which indicates that the dipole is close to resonance there when the switch is closed, while the open switch detunes the antenna and suppresses its scattering. The CST analysis thus confirms good distinction between the ON and OFF states of the designed tag.

6.4 GNU Radio Signal-Level Results

6.4.1 Power and SNR Measurement

The power-estimation branches square the relevant signals and average them with moving-average blocks, and the SNR is computed as in Equation (5.5). The values obtained in the baseline run are collected in Figure 6.1. The backscatter signal power is 2.1×10^{-5} in the units of the flowgraph. The noise power at the baseline noise setting is below 10^{-6}, the six-decimal resolution of the number sink, so it reads as 0.000000. These values are relative quantities inside the model; they set the operating point of the simulation rather than an absolute RF power.

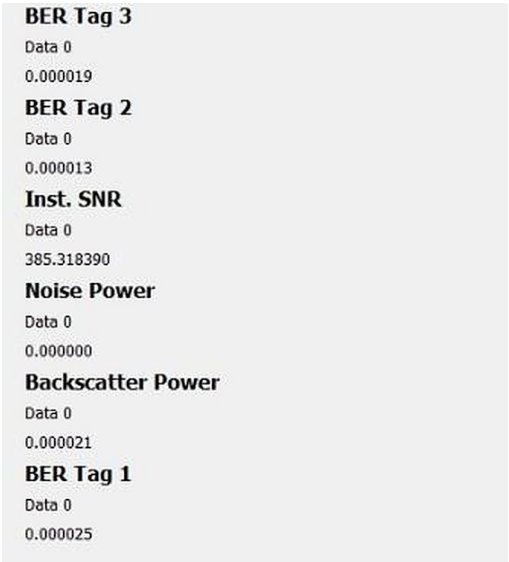


Figure 6.1 – Number Sink outputs of the baseline simulation: BER of Tags 1–3, backscatter power, noise power and instantaneous SNR

6.4.2 Received Spectrum

The received spectra were recorded with the Qt GUI Frequency Sink (50 kHz bandwidth, 32 768-point FFT) after the band-pass filters of each branch. Because the signals are real-valued, every component appears at both positive and negative frequencies. The subcarrier lines lie at ± 11 kHz for Tag 1, ± 10.8 kHz for Tag 2 and ± 11.2 kHz for Tag 3 (Figures 6.2–6.4), which are the 10 kHz carrier offset by each tag's control frequency of 1 kHz, 800 Hz and 1.2 kHz. These positions agree with the subcarrier relation $F_c \pm F_{tag}$ of the base thesis [1]. The cascaded 160 Hz filters attenuate everything outside the selected band, so each branch passes essentially only its own tag. The Tag 1 spectrum also contains lines at ± 10 kHz, the residual carrier, while the Tag 2 and Tag 3 spectra contain only weak traces near ± 10 kHz.

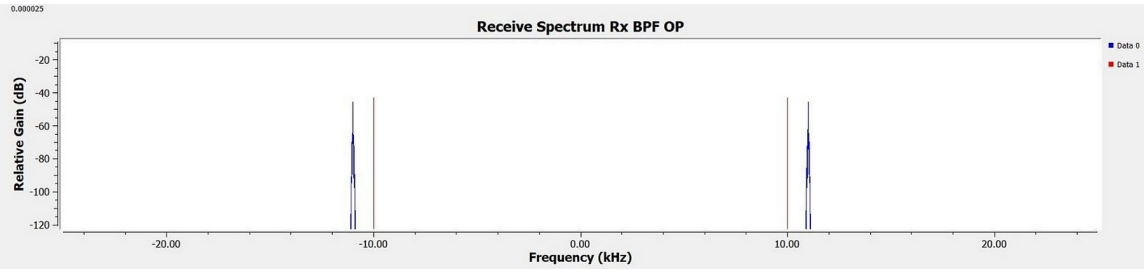


Figure 6.2 – Received spectrum of the Tag 1 branch with lines at ± 11 kHz and at the ± 10 kHz carrier frequency

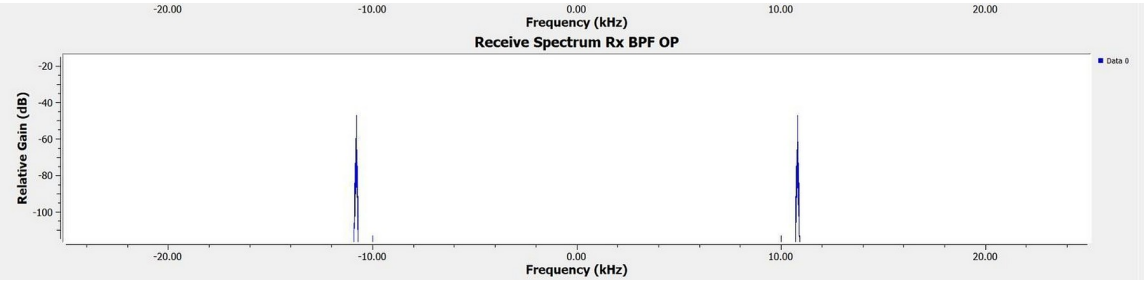


Figure 6.3 – Received spectrum of the Tag 2 branch after band-pass filtering, with lines at ± 10.8 kHz

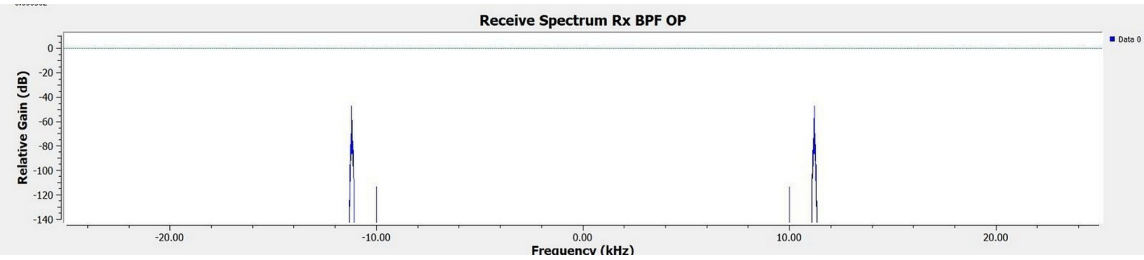


Figure 6.4 – Received spectrum of the Tag 3 branch after band-pass filtering, with lines at ± 11.2 kHz

6.4.3 Time-Domain Signals and Delay Alignment

The Qt GUI Time Sinks recorded, for each tag, the original baseband data, the envelope-detector output, the low-pass filter output, the matched-filter output and the AGC output (Figures 6.5–6.7). The horizontal axis is time in milliseconds and the vertical axis is amplitude. The original data form a rectangular bit pattern, and the AGC output follows the same pattern after processing, delayed by the receiver chain. The 16 000-sample delay aligns the reference sequence with the detected signal, and the further 1250-sample delay places the reference sampling point at the middle of each bit. These traces were used to set the alignment before the BER was recorded.

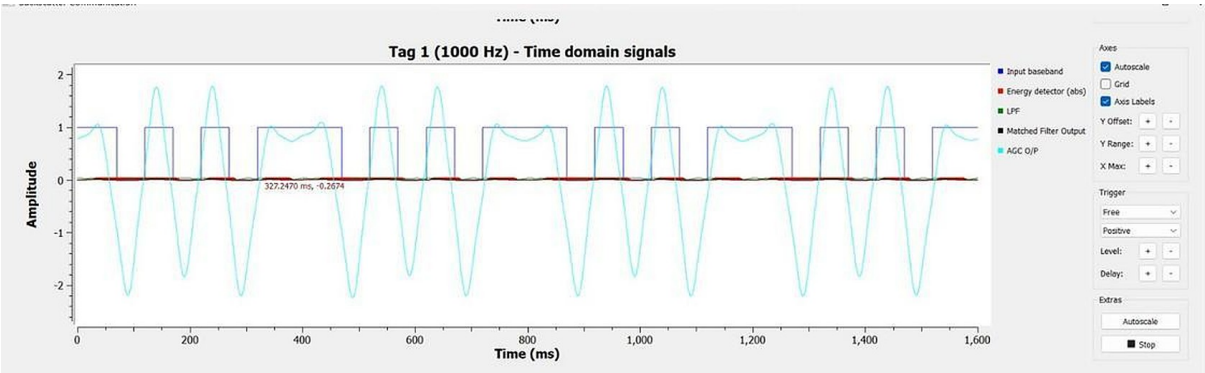


Figure 6.5 – Time-domain signals of the Tag 1 branch (control frequency 1000 Hz): input data and receiver-stage outputs

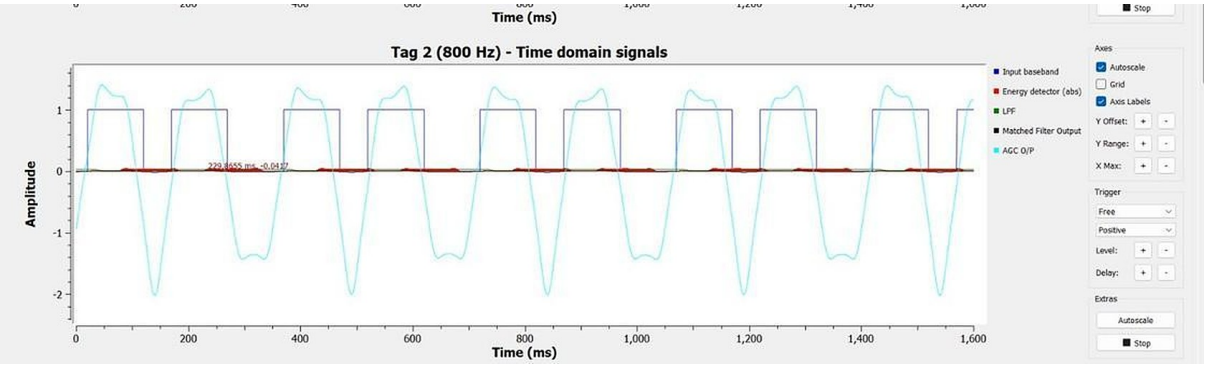


Figure 6.6 – Time-domain signals of the Tag 2 branch (control frequency 800 Hz)

6.4.4 Recovered Symbol Outputs

The recovered symbols at the Symbol Sync outputs of the three branches are plotted in Figure 6.8. The horizontal axis is time in seconds and the vertical axis is the amplitude of each symbol sample. Each vertical line is one sample taken at the recovered symbol instant rather than part of a continuous waveform. The window holds 64 points at 20 symbols/s, which spans 3.2 s. The two symbol levels are clearly separated in all three branches, and the Binary Slicer

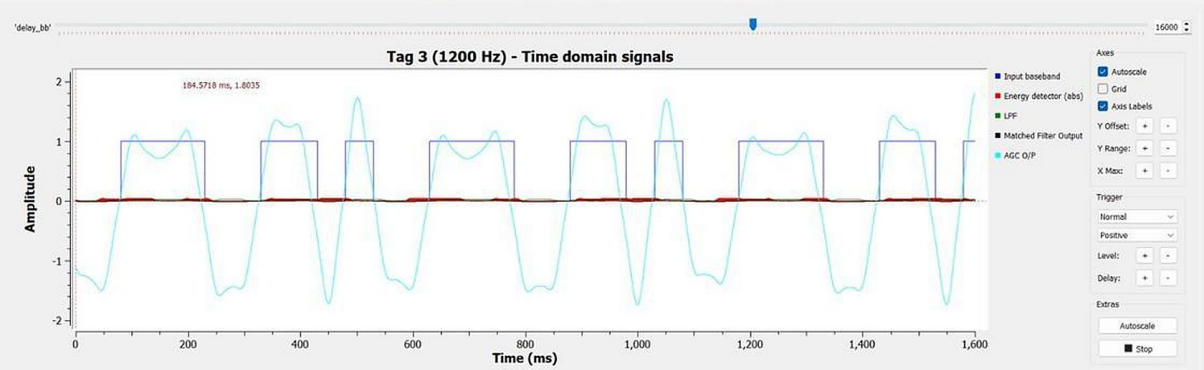


Figure 6.7 – Time-domain signals of the Tag 3 branch (control frequency 1200 Hz) with the delay-alignment control

converts the sign of each sample into a bit. The differences in amplitude between the tags follow from the different signal content of each branch after filtering and AGC.

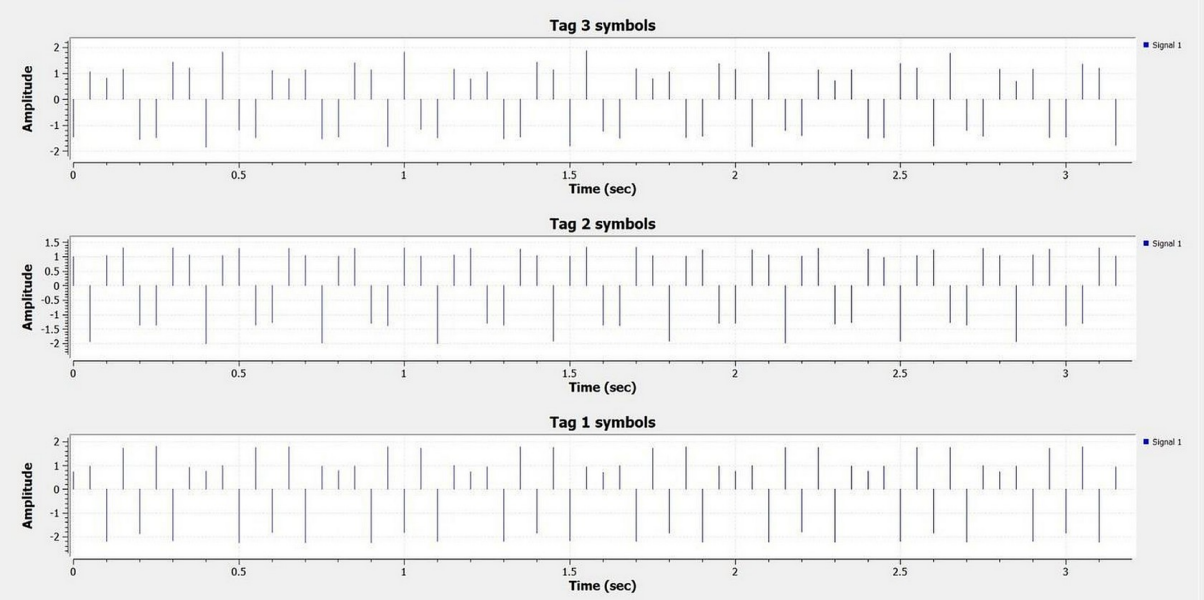


Figure 6.8 – Recovered symbol samples at the output of the Symbol Sync block for Tags 1, 2 and 3

6.4.5 BER Results

The BER values obtained in the baseline run are listed in Table 6.2. All three tags were recovered with BER values in the order of 10^{-5}, with Tag 2 lowest at 1.3×10^{-5}. The noise sweep results are listed in Table 6.3 and plotted in Figure 6.9.

The BER rises monotonically with the noise power for every tag. Up to a noise setting of 2×10^{-5} all tags stay at or below about 10^{-3}. Between 2×10^{-5} and 3×10^{-5} the BER increases by more than an order of magnitude, which marks the point at which noise begins to cross the decision threshold regularly. At the highest setting all three tags approach a BER of 0.5, the

Table 6.2 – BER and power values obtained in the baseline simulation

Quantity	Obtained value
BER, Tag 1	0.000025
BER, Tag 2	0.000013
BER, Tag 3	0.000019
Backscatter power	0.000021
Noise power	< 0.000001

Table 6.3 – BER of the three tags for different noise-power settings

Noise power (input value)	Inst. SNR (dB)	BER Tag 1	BER Tag 2	BER Tag 3
0.000010	−3.758240	0.000362	0.000091	0.000136
0.000020	−3.743252	0.001128	0.000376	0.000564
0.000030	−9.769442	0.025215	0.015021	0.019313
0.000050	−13.293872	0.078824	0.061765	0.071176
0.000080	−17.720554	0.230219	0.220588	0.239057
0.000100	−21.803251	0.387438	0.361318	0.381219
0.000200	−29.745438	0.476648	0.481685	0.484890

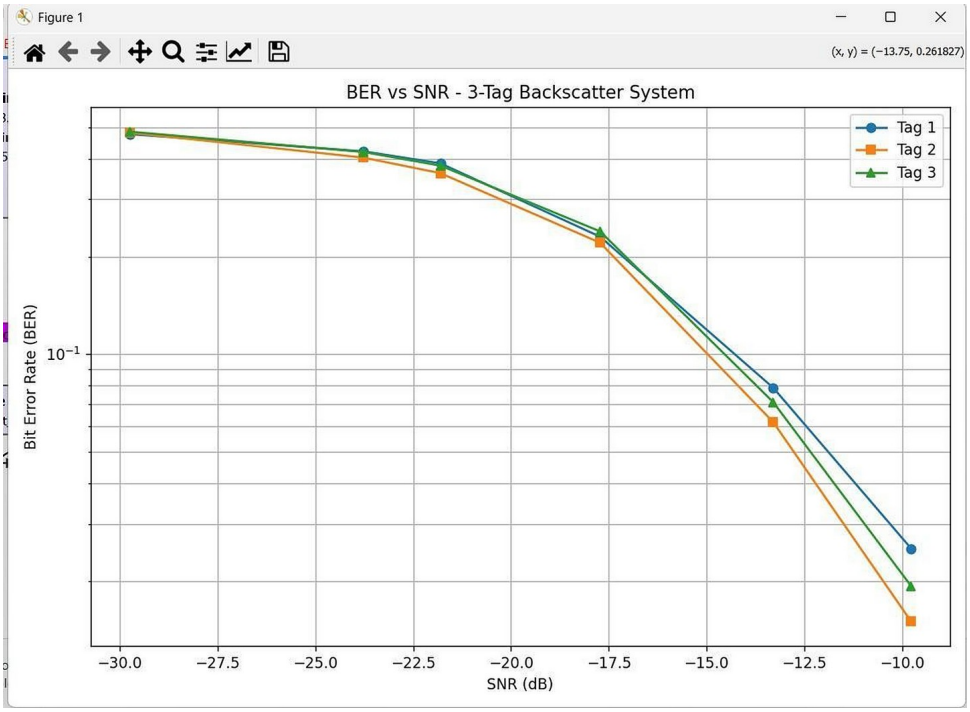


Figure 6.9 – BER versus SNR for the three-tag backscatter system obtained from the GNU Radio model

value expected when the decisions are no better than random. The SNR values in the table are instantaneous readings taken at each setting; at the two lowest noise settings the readings are close to each other, while the BER still separates the two cases clearly. The three tags follow the same trend, with Tag 2 slightly better than the others at the lower noise levels. Since all branches share the same processing, this small difference comes from the different data sequences and subcarrier positions of the tags. The results confirm that the receiver separates and recovers three simultaneously active subcarrier tags from one composite signal, and that its performance degrades gradually as the noise increases.

6.5 Throughput Evaluation of the Standalone IRSA–SIC Simulation

Figure 6.10 presents the output of the project code for IRSA with SIC, with the PLR on a logarithmic scale on the left and the throughput on the right, both against the channel load G. The simulation conditions are $N = 200$ slots, $\Lambda(x) = 0.5x^2 + 0.28x^3 + 0.22x^8$, 800 frames per load point, a collision channel and ideal SIC. The dashed line marks the ideal throughput $T = G$, and the marked point is the peak throughput.

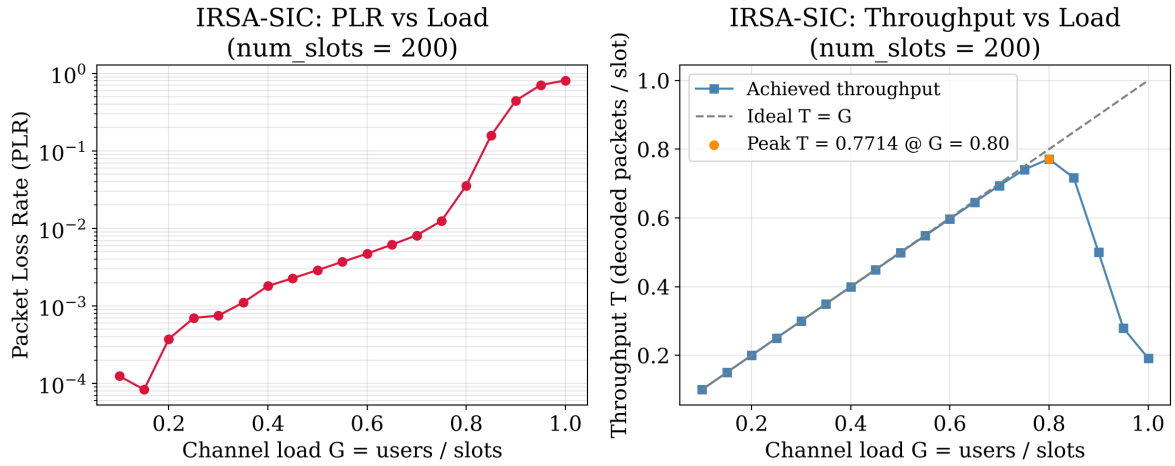


Figure 6.10 – IRSA with SIC: packet loss rate (left) and throughput (right) versus channel load G ($N = 200$ slots, $\Lambda(x) = 0.5x^2 + 0.28x^3 + 0.22x^8$, 800 frames per point)

The peak throughput is 0.7714 decoded packets per slot, reached at $G = 0.80$, where 160 users contend for 200 slots and the PLR is 3.6%. Up to $G = 0.75$ the throughput stays within 1.3% of the ideal line, and the PLR remains between 10^{-4} and about 10^{-2}. Beyond $G = 0.80$ the throughput falls steeply: 0.716 at $G = 0.85$, 0.500 at $G = 0.90$ and 0.192 at $G = 1.00$, where 81% of the users are lost.

6.6 Comparison with Baseline Methods

Table 6.4 lists the throughput of all four schemes, together with the IRSA PLR, at every load point. All values were obtained with the same frame size, load points, trials and seed.

Table 6.4 – Throughput (packets/slot) of IRSA with SIC and the baseline schemes under identical conditions ($N = 200$, 800 frames per load)

G	M	IRSA + SIC	IRSA PLR	CRDSA + SIC	FSA	IRSA, no SIC
0.10	20	0.1000	1.25×10^{-4}	0.0998	0.0907	0.0950
0.15	30	0.1500	8.33×10^{-5}	0.1496	0.1299	0.1344
0.20	40	0.1999	3.75×10^{-4}	0.1992	0.1651	0.1663
0.25	50	0.2498	7.00×10^{-4}	0.2486	0.1959	0.1905
0.30	60	0.2998	7.50×10^{-4}	0.2979	0.2224	0.2077
0.35	70	0.3496	1.11×10^{-3}	0.3464	0.2470	0.2191
0.40	80	0.3993	1.81×10^{-3}	0.3933	0.2674	0.2235
0.45	90	0.4490	2.28×10^{-3}	0.4384	0.2867	0.2234
0.50	100	0.4985	2.90×10^{-3}	0.4793	0.3035	0.2182
0.55	110	0.5480	3.70×10^{-3}	0.5137	0.3174	0.2097
0.60	120	0.5972	4.73×10^{-3}	0.5334	0.3295	0.1983
0.65	130	0.6460	6.17×10^{-3}	**0.5370**	0.3395	0.1858
0.70	140	0.6943	8.15×10^{-3}	0.5232	0.3482	0.1710
0.75	150	0.7406	1.25×10^{-2}	0.5016	0.3541	0.1569
0.80	160	**0.7714**	3.57×10^{-2}	0.4730	0.3596	0.1423
0.85	170	0.7164	1.57×10^{-1}	0.4448	0.3632	0.1285
0.90	180	0.4998	4.45×10^{-1}	0.4148	0.3657	0.1154
0.95	190	0.2790	7.06×10^{-1}	0.3891	0.3674	0.1031
1.00	200	0.1916	8.08×10^{-1}	0.3650	**0.3678**	0.0922

The throughput curves of all schemes, together with the analytical slotted ALOHA curve Ge^{-G}, are plotted in Figure 6.11, and the corresponding PLR curves in Figure 6.12. Table 6.5 summarises the key figures.

The FSA results agree with the analytical curve Ge^{-G} across the whole range, peaking at 0.3678 at $G = 1.0$, which also confirms the correctness of the simulator. CRDSA reaches 0.5370 at $G = 0.65$, close to the value of about 0.55 reported by Casini *et al.* [18]. IRSA with SIC reaches 0.7714, which is 2.10 times the FSA peak and 1.44 times the CRDSA peak.

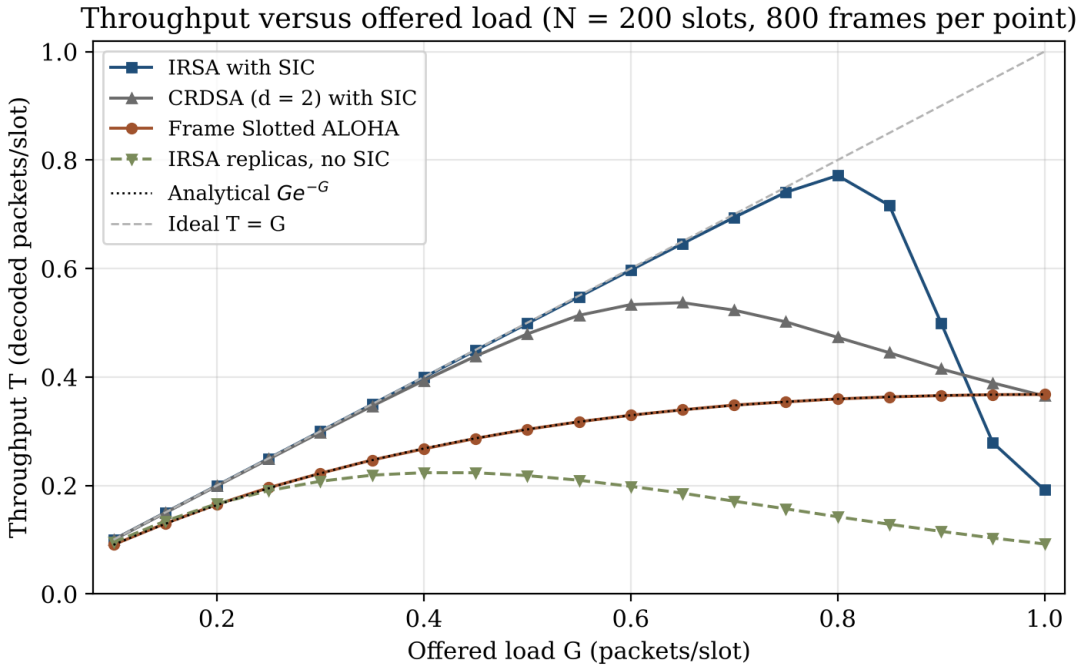


Figure 6.11 – Throughput versus offered load for IRSA with SIC, CRDSA with SIC, Frame Slotted ALOHA and IRSA without SIC ($N = 200$ slots, 800 frames per point)

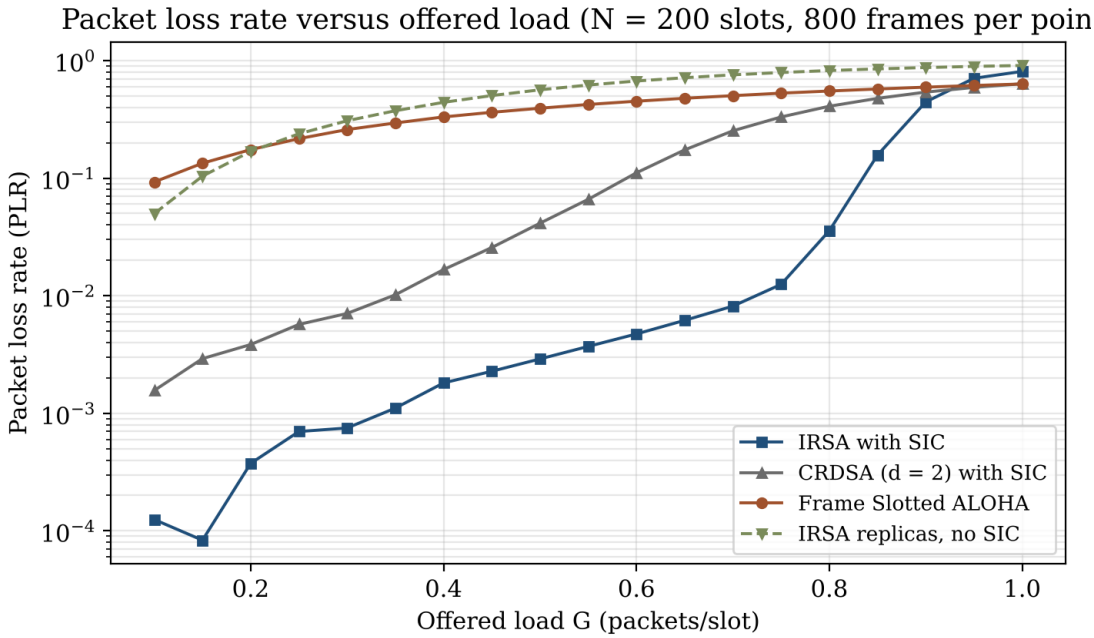


Figure 6.12 – Packet loss rate versus offered load for the four schemes ($N = 200$ slots, 800 frames per point)

Table 6.5 – Summary of simulation parameters and key throughput results

Scheme	Degree distribution	Peak T (packets/slot)	Load at peak	PLR at $G = 0.5$
IRSA + SIC	$0.5x^2 + 0.28x^3 + 0.22x^8$	0.7714	0.80	2.9×10^{-3}
CRDSA + SIC	x^2	0.5370	0.65	4.1×10^{-2}
FSA	x	0.3678	1.00	3.9×10^{-1}
IRSA, no SIC	$0.5x^2 + 0.28x^3 + 0.22x^8$	0.2235	0.40	5.6×10^{-1}

Common conditions: $N = 200$ slots; $G = 0.10$–1.00 in steps of 0.05; 800 frames per load; master seed 0; collision channel without capture; ideal cancellation.

6.7 Discussion of the Throughput Behaviour

Low and moderate load. For $G \leq 0.75$ the IRSA throughput tracks the ideal line $T = G$, so nearly every user is recovered. At $G = 0.5$, for example, 100 users send on average 360 replicas into 200 slots, so most slots are collisions, yet SIC recovers 99.7% of the users. Over the same range FSA loses between 9% and 53% of its packets, and CRDSA departs from the ideal line above $G \approx 0.5$.

Error floor. At low load the IRSA PLR does not fall towards zero but settles between about 10^{-4} and 10^{-3}. This floor comes from the stopping sets described in Sections 2.13 and 5.5.3: when two degree-2 users choose the same pair of slots, neither ever becomes a singleton. The probability of such a pair falls as the frame grows, so the floor is a finite-frame effect. The irregularity of the distribution also explains why the floor is low: half of the users have degree 2 and are prone to such loops, but the degree-8 users are almost always decoded early and clear many slots.

Threshold behaviour. Between $G = 0.80$ and $G = 0.90$ the IRSA throughput falls from 0.7714 to 0.500, and the PLR rises from 3.6% to 44.5%. This sharp transition is the finite-frame form of the decoding threshold predicted by the graph analysis of IRSA [19]. The asymptotic threshold of the chosen distribution is about 0.94; with 200 slots, the transition begins earlier and is spread over a range of loads, so the useful operating limit is close to $G = 0.8$. Above $G \approx 0.92$ the IRSA throughput falls below that of CRDSA, and soon after below that of FSA. At these loads the 3.6 replicas per user fill the frame so densely (more than 3.3 replicas per slot) that the SIC process stalls almost at the start, and the replicas themselves become the main source of interference.

Role of repetition and SIC. The IRSA-without-SIC curve isolates the two ingredients. With repetition alone, the throughput peaks at only 0.2235 at $G = 0.40$, which is lower than FSA at every load from $G = 0.25$ upward. Repetition without cancellation simply multiplies the

number of colliding transmissions. Conversely, cancellation has no effect in FSA, where each user has only one transmission. The large gain of IRSA therefore comes from the combination: repetition creates multiple opportunities to find a clean copy, and SIC turns each decoded copy into the removal of interference from other slots. The comparison of IRSA with CRDSA reveals the further value of irregularity, since a mixture of low-degree and high-degree users sustains the iterative decoding up to a much higher load than a fixed degree of two.

Cost of the gain. The throughput gain is obtained at the expense of transmitted energy and slot occupancy. Each IRSA user transmits 3.6 times on average, compared with once in FSA and twice in CRDSA, so a battery-assisted tag spends correspondingly more switching energy per delivered packet. For backscatter tags this cost is small, because the switching energy of a device such as the ADG902 is in the nanojoule range [1], but it should be weighed in tags powered solely by harvested energy. A further practical consideration is that the gain relies on the reader being able to reconstruct and cancel replicas accurately. Residual interference after imperfect cancellation, and capture of a strong packet in a collided slot, would change these curves in practice, and they form part of the future work.

Operating point for multi-tag backscatter. The results suggest that a reader using IRSA with SIC can serve up to about 0.75 tags per slot with a packet loss rate near 1%, compared with a peak of 0.37 tags per slot for the framed ALOHA procedure used in conventional RFID. Because the decline above the threshold is steep, a practical system would benefit from estimating the number of active tags and adapting the frame length so that the load stays below about 0.8, in the same spirit as the adaptive frame length of Schoute [16].

6.8 Discussion: Objectives, Advantages and Limitations

6.8.1 Achievement of the Objectives

The project set out five objectives in Section 1.5, covering the study of backscatter principles, the tag hardware design, the electromagnetic analysis, the signal-level model and the packet-level multiple-access evaluation. Each objective was addressed in its own workstream, and each produced a concrete, reproducible outcome rather than a purely descriptive one. The study of the base thesis and the multiple-access literature fixed the design choices used later: the ADG902 switch and the separate controller board follow the tag architecture of the base work, the three-subcarrier GNU Radio model applies its frequency-division principle, and the IRSA/SIC simulation takes up the time-domain multiple access that the base work leaves as future work. Table 6.6 relates each objective to the corresponding result.

The outcomes also show how far each objective has been taken. The hardware and electromagnetic objectives were completed at the design and simulation stage, with fabrication and

measurement planned next. The signal-level objective was met with a working three-tag receiver and a full BER-versus-SNR characterisation. The packet-level objective was met with a verified simulation and a comparison against three baselines under identical conditions. Together, these results provide a consistent basis for the integration work described in Chapter 7.

Table 6.6 – Achievement of the project objectives

No.	Objective	Outcome
1	Study backscatter principles	Reviewed load modulation, RCS, link budget and receiver processing, with a detailed review of the base thesis (Chapter 2)
2	Design the tag PCB in KiCad	Schematic, routed layout and 3D model completed with a four-pin interface to an external ESP32-S3
3	Analyse the tag in CST	RCS contrast of about 18.2 dB between ON and OFF states at 220 MHz
4	Develop the GNU Radio model	Three subcarrier tags separated and recovered; BER of order 10^{-5} at baseline; BER-versus-SNR obtained
5	Implement and evaluate IRSA/SIC	Peak throughput 0.7714 at $G = 0.80$, against 0.5370 for CRDSA and 0.3678 for FSA under identical conditions

6.8.2 Advantages of the Proposed Approach

The tag modulates its reflection with a single reflective switch instead of an active RF chain, so the architecture may reduce the power requirement of the RF front-end; the actual consumption of the tag remains to be measured once the board is fabricated. The separate controller board also keeps the tag simple and reusable. The CST analysis confirms that the chosen switch states produce a strong scattering contrast. The GNU Radio model demonstrates that subcarrier separation and a simple non-coherent receiver are sufficient to recover several simultaneously active tags. At packet level, IRSA with SIC more than doubles the throughput of framed ALOHA without requiring any coordination among tags. The tags only need to choose random slots, which suits the limited capabilities of backscatter devices.

6.8.3 Limitations of the Present Results

The tag was not fabricated within the project period, so its electrical behaviour and the RCS contrast have not yet been confirmed by measurement. The GNU Radio model uses a baseband-equivalent carrier, numerical response coefficients and additive Gaussian noise, and does not include multipath or frequency offset. The IRSA/SIC simulation assumes a collision channel with ideal cancellation and was evaluated separately from the signal model and the hardware. These limitations define the next steps described in Chapter 7.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Summary of the Completed Work

Backscatter communication enables a device to send data by controlling how it reflects an incident wave, so it needs no active transmitter and can operate at very low power and cost. Its challenges are a weak reflected signal, strong interference from the illuminating carrier and collisions when many tags share a reader. This project addressed these challenges through four workstreams.

In the **hardware workstream**, a switch-loaded tag was designed in KiCad. The tag PCB carries the ADG902 RF switch, a TPS7A20 regulator for the 2.5 V rail, an SN74LVC1T45 translator for the control logic and the antenna pads, and it connects to an external ESP32-S3 controller through a four-pin connector.

In the **electromagnetic workstream**, the tag design was imported into CST Studio Suite and its RCS was computed for the ON and OFF switch conditions. The two states differ by about 18.2 dB at 220 MHz, which confirms a clear distinction between the backscatter states.

In the **signal-level workstream**, a three-tag model was built in GNU Radio Companion. The tags were separated by their subcarriers and recovered by a band-pass, envelope-detection and matched-filter receiver, with BER values of 0.000025, 0.000013 and 0.000019 in the baseline run and a consistent BER-versus-SNR trend under a noise sweep.

In the packet-level workstream, IRSA with SIC was implemented independently in Python and evaluated by Monte Carlo simulation with 200 slots per frame. The scheme reached a peak throughput of 0.7714 packets per slot at a load of 0.80, compared with 0.5370 for CRDSA and 0.3678 for Frame Slotted ALOHA under identical conditions. This simulation was carried out separately and was not integrated with the backscatter hardware or the GNU Radio model.

7.2 Major Findings

1. A reflective SPST switch with a low ON resistance and a small OFF capacitance gives the tag a strong scattering contrast of about 18.2 dB at 220 MHz, corresponding to a 66-fold change in backscattered power between the two states.
2. Subcarrier separation, the multiple-access principle of the base thesis, allows three tags transmitting simultaneously to be recovered from one composite signal with BER values of order 10^{-5} using a simple non-coherent receiver.
3. IRSA with SIC more than doubles the peak throughput of framed ALOHA and maintains a packet loss rate near 1% up to a load of 0.75 packets per slot.
4. Repetition without cancellation lowers the throughput below that of plain ALOHA; the gain of IRSA comes from the combination of irregular repetition and iterative cancellation.
5. The IRSA throughput collapses sharply above a load of about 0.8 for a 200-slot frame, so a practical reader should keep the load below this point, for example by adapting the frame length.

7.3 Major Contributions of the Project

The project brings together four complementary pieces of work on multi-tag backscatter communication. It contributes a tag design that cleanly separates the RF switching board from the controller; an electromagnetic analysis that quantifies the state contrast of that design; a signal-level model that demonstrates subcarrier-based separation and recovery of several tags; and an independent packet-level implementation of IRSA with SIC, verified against the analytical ALOHA result and compared with CRDSA, FSA and a no-SIC reference under identical conditions. Taken together, they cover the physical source of the modulation, its detection at signal level and its sharing among many tags at packet level, and they extend the multi-tag question that the base thesis left as future work.

7.4 Overall Conclusion

All five objectives of Section 1.5 were met (Table 6.6). The work shows that the elements required for a scalable multi-tag backscatter system are each achievable with simple means: a single-switch tag with a strong state contrast, a non-coherent receiver that separates tags by subcarrier, and a random-access scheme that serves about twice as many tags per slot as framed ALOHA without any coordination among the tags. Each element is established in its own workstream, which provides a clear foundation for combining them.

7.5 Constraints in the Current Work

The tag was designed and analysed but could not be fabricated within the project period because of time constraints, so the hardware and its RCS contrast have not yet been tested physically. The GNU Radio model represents the link with a baseband-equivalent carrier and additive Gaussian noise and does not include multipath, frequency offset or phase noise. The IRSA/SIC study has been implemented and evaluated independently at packet level under a collision channel with ideal cancellation; its integration with the GNU Radio signal model and the physical backscatter system has not yet been completed.

7.6 Future Scope

7.6.1 PCB Fabrication and Hardware Testing

The first step is to fabricate and assemble the tag from the KiCad design. The regulator output, the translator levels and the switching of the ADG902 can then be checked with the ESP32-S3 connected through J1, before the antenna is attached.

7.6.2 Integration with the Backscatter RF Set-Up

The assembled tag can be placed in a bistatic set-up with an RF source and an SDR reader, as in the base thesis. Measuring the received power in the two switch states will allow the CST contrast and the link-budget relation of Equation (2.4) to be compared with the physical link, and will reveal effects such as component tolerances, cable coupling and environmental reflections.

7.6.3 Validation and Extension of the GNU Radio Model

The GNU Radio receiver can be connected to SDR captures of the real tag in place of the simulated composite signal. The model itself can be extended with carrier frequency offset, multipath and a packet structure with a preamble, so that it produces per-packet decoding decisions in addition to bit decisions.

7.6.4 Integration of IRSA and SIC with the Backscatter System

The IRSA/SIC implementation already exists and has been evaluated at packet level, so integration does not require a new implementation. The next step is to connect it to the lower layers. The ESP32-S3 firmware can be extended to transmit packet replicas in randomly chosen slots of a frame announced by the reader. The signal-level receiver can then decide, for each slot, whether a clean packet is present, and feed this slot information to the existing decoding routine. The packet model can be refined with imperfect cancellation, in which a fraction of each cancelled replica remains as interference, and with capture, in which a strong packet survives a collision, so that its predictions reflect the behaviour of a real reader.

7.6.5 Evaluation of Multiple-Access Performance

With the integrated system, throughput, packet loss rate and energy per delivered packet can be measured against the offered load and compared with the simulated curves of Chapter 6. Other repetition-degree distributions, different frame sizes and adaptive frame-length control can be evaluated to find the best operating point for backscatter tags.

7.6.6 Physical Multi-Tag Demonstration

A demonstration with several fabricated tags sharing one reader would establish the relationship between the packet-level simulation and the actual backscatter system: the CST and RF measurements would confirm the per-tag modulation, the SDR receiver would confirm per-slot decoding, and the IRSA/SIC procedure applied to the received slots would confirm the throughput gain under real propagation and imperfect cancellation.

7.7 Potential Applications

Multi-tag backscatter with efficient random access suits applications in which many simple nodes report small amounts of data: precision agriculture with leaf, soil and microclimate sensors, as targeted by the base thesis; inventory and asset tracking in warehouses; structural and environmental monitoring in buildings; and battery-free wearable or body-area sensors.

7.8 Final Remarks

By combining PCB design, electromagnetic analysis, signal-level modelling and packet-level multiple-access simulation, the project establishes a base for multi-tag backscatter communication. Fabricating the tag, validating the link over the air and integrating IRSA with SIC into the receiver chain are the natural next stages of this work.

REFERENCES

[1] S. N. Daskalakis, "Ambient backscatterers for low cost and low power wireless applications," Ph.D. dissertation, Inst. Signals, Sensors Syst., Heriot-Watt Univ., Edinburgh, U.K., Feb. 2020.

[2] S. N. Daskalakis, J. Kimionis, A. Collado, G. Goussetis, M. M. Tentzeris, and A. Georgiadis, "Ambient backscatterers using FM broadcasting for low cost and low power wireless applications," *IEEE Trans. Microw. Theory Techn.*, vol. 65, no. 12, pp. 5251–5262, Dec. 2017.

[3] S. N. Daskalakis, G. Goussetis, S. D. Assimonis, M. M. Tentzeris, and A. Georgiadis, "A uW backscatter-Morse-leaf sensor for low-power agricultural wireless sensor networks," *IEEE Sensors J.*, vol. 18, no. 19, pp. 7889–7898, Oct. 2018.

[4] S. N. Daskalakis, R. Correia, G. Goussetis, M. M. Tentzeris, N. B. Carvalho, and A. Georgiadis, "Four-PAM modulation of ambient FM backscattering for spectrally efficient low-power applications," *IEEE Trans. Microw. Theory Techn.*, vol. 66, no. 12, pp. 5909–5921, Dec. 2018.

[5] D. Belo, R. Correia, Y. Ding, S. N. Daskalakis, G. Goussetis, A. Georgiadis, and N. B. Carvalho, "IQ impedance modulator front-end for low-power LoRa backscattering devices," *IEEE Trans. Microw. Theory Techn.*, 2019.

[6] V. Liu, A. Parks, V. Talla, S. Gollakota, D. Wetherall, and J. R. Smith, "Ambient backscatter: Wireless communication out of thin air," in *Proc. ACM SIGCOMM*, 2013, pp. 39–50.

[7] N. Van Huynh, D. T. Hoang, X. Lu, D. Niyato, P. Wang, and D. I. Kim, "Ambient backscatter communications: A contemporary survey," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 2889–2922, 4th Quart. 2018.

[8] C. Boyer and S. Roy, "Backscatter communication and RFID: Coding, energy, and MIMO analysis," *IEEE Trans. Commun.*, vol. 62, no. 3, pp. 770–785, Mar. 2014.

[9] J. Kimionis, A. Bletsas, and J. N. Sahalos, "Increased range bistatic scatter radio," *IEEE Trans. Commun.*, vol. 62, no. 3, pp. 1091–1104, Mar. 2014.

[10] P. V. Nikitin and K. V. S. Rao, "Theory and measurement of backscattering from RFID tags," *IEEE Antennas Propag. Mag.*, vol. 48, no. 6, pp. 212–218, Dec. 2006.

[11] J. D. Griffin and G. D. Durgin, "Complete link budgets for backscatter-radio and RFID systems," *IEEE Antennas Propag. Mag.*, vol. 51, no. 2, pp. 11–25, Apr. 2009.

[12] K. Finkenzeller, *RFID Handbook: Fundamentals and Applications in Contactless Smart Cards, Radio Frequency Identification and Near-Field Communication*, 3rd ed. Chichester, U.K.: Wiley, 2010.

[13] J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed. New York, NY, USA: McGraw-Hill, 2008.

[14] N. Abramson, "The ALOHA system: Another alternative for computer communications," in *Proc. AFIPS Fall Joint Comput. Conf.*, 1970, pp. 281–285.

[15] L. G. Roberts, "ALOHA packet system with and without slots and capture," *ACM SIGCOMM Comput. Commun. Rev.*, vol. 5, no. 2, pp. 28–42, Apr. 1975.

[16] F. C. Schoute, “Dynamic frame length ALOHA,” *IEEE Trans. Commun.*, vol. 31, no. 4, pp. 565–568, Apr. 1983.

[17] EPCglobal Inc., “EPC radio-frequency identity protocols Class-1 Generation-2 UHF RFID protocol for communications at 860 MHz–960 MHz,” Version 1.2.0, 2008.

[18] E. Casini, R. De Gaudenzi, and O. del Rio Herrero, “Contention resolution diversity slotted ALOHA (CRDSA): An enhanced random access scheme for satellite access packet networks,” *IEEE Trans. Wireless Commun.*, vol. 6, no. 4, pp. 1408–1419, Apr. 2007.

[19] G. Liva, “Graph-based analysis and optimization of contention resolution diversity slotted ALOHA,” *IEEE Trans. Commun.*, vol. 59, no. 2, pp. 477–487, Feb. 2011.

[20] E. Paolini, G. Liva, and M. Chiani, “Coded slotted ALOHA: A graph-based method for uncoordinated multiple access,” *IEEE Trans. Inf. Theory*, vol. 61, no. 12, pp. 6815–6832, Dec. 2015.

[21] GNU Radio Project, “GNU Radio: The free and open software radio ecosystem.” [Online]. Available: https://www.gnuradio.org

[22] C. R. Harris *et al.*, “Array programming with NumPy,” *Nature*, vol. 585, pp. 357–362, Sep. 2020.

[23] J. D. Hunter, “Matplotlib: A 2D graphics environment,” *Comput. Sci. Eng.*, vol. 9, no. 3, pp. 90–95, 2007.

[24] KiCad EDA, “KiCad: A cross-platform and open-source electronics design automation suite.” [Online]. Available: https://www.kicad.org

[25] Dassault Systèmes, “CST Studio Suite: Electromagnetic field simulation software.” [Online]. Available: https://www.3ds.com/products/simulia/cst-studio-suite

[26] Analog Devices, “ADG901/ADG902: Wideband 2.5 GHz, 37 dB isolation at 1 GHz, CMOS 1.65 V to 2.75 V, SPST switches,” data sheet.

[27] Texas Instruments, “TPS7A20 300-mA, ultra-low-noise, low-IQ, high-PSRR low-dropout (LDO) linear regulator,” data sheet.

[28] Texas Instruments, “SN74LVC1T45 single-bit dual-supply bus transceiver with configurable voltage translation and 3-state outputs,” data sheet.

[29] Espressif Systems, “ESP32-S3 series datasheet,” and “ESP32-S3-WROOM-1 datasheet.”

APPENDICES

A Summary of Evidence Level

Table A.1 summarises the evidence available for each element of the project, as reported in Chapters 3 to 6.

Table A.1 – Evidence level of the project elements

Project element	Reported evidence	Status
Tag schematic and PCB layout (KiCad)	Schematic, routed layout and 3D model (Figures 4.1, 5.1); external ESP32-S3 interface through J1 (Figure 4.2)	Design completed; fabrication pending
RCS of the tag (CST)	RCS versus frequency for ON and OFF loads; 18.2 dB contrast at 220 MHz (Figure 5.3, Table 6.1)	Simulation completed; measurement pending
Three-tag GNU Radio model	Spectra, time-domain signals, recovered symbols, baseline BER and BER versus SNR (Figures 6.2–6.9, Tables 6.2, 6.3)	Signal-level simulation completed
IRSA and SIC	Standalone Python packet-level simulation with throughput and PLR evaluation; FSA, CRDSA and no-SIC baselines (Figures 6.10–6.12, Tables 6.4, 6.5)	Simulation completed; integration with GNU Radio and physical hardware pending
Over-the-air backscatter link	–	Planned as future work

B Commands to Reproduce the Results

The packet-level results can be regenerated with Python 3 and the NumPy and Matplotlib packages:

```
pip install numpy matplotlib
python irsa_sic_main.py      # Listing C.1: IRSA-SIC sweep, saves irsa_sic_result.png
python baselines.py          # Listing C.2: FSA, CRDSA and IRSA without SIC
```

For Listing C.2, the four functions of Listing C.1 (`build_transmit_matrix`, `irsa_sic_decode`, `simulate_frame` and `monte_carlo`) are saved as `irsa_sic.py` in the same folder. All runs use the master seed 0, so the printed values match Table 6.4.

The GNU Radio results are regenerated by opening the project flowgraph in GNU Radio Companion and running it with the parameters of Table 4.4; the BER-versus-SNR data are produced by changing the noise amplitude to each value in Table 6.3.

C Source Listings

```
import numpy as np
import matplotlib.pyplot as plt


def build_transmit_matrix(num_users, num_slots, degree_dist, rng):
    degrees = np.array(list(degree_dist.keys()))
    probs = np.array(list(degree_dist.values()))
    probs = probs / probs.sum()

    M = np.zeros((num_users, num_slots), dtype=np.int8)

    for u in range(num_users):
        d = rng.choice(degrees, p=probs)
        d = min(d, num_slots)
        slots = rng.choice(num_slots, size=d, replace=False)
        M[u, slots] = 1

    return M


def irsa_sic_decode(M):
    M = M.copy()
    num_users, num_slots = M.shape
    decoded_users = []

    while True:
        col_sums = M.sum(axis=0)
        singleton_slots = np.where(col_sums == 1)[0]

        if len(singleton_slots) == 0:
            break

        newly_decoded = set()
        for s in singleton_slots:
            user_rows = np.where(M[:, s] == 1)[0]
            if len(user_rows) != 1:
                continue
            u = user_rows[0]
            newly_decoded.add(u)

        if not newly_decoded:
            break

        for u in newly_decoded:
            decoded_users.append(u)
            M[u, :] = 0

```

```
        if len(decoded_users) == num_users:
            break

    return sorted(decoded_users)


def simulate_frame(num_users, num_slots, degree_dist, seed=None):
    rng = np.random.default_rng(seed)
    M = build_transmit_matrix(num_users, num_slots, degree_dist, rng)
    decoded_users = irsa_sic_decode(M)

    num_decoded = len(decoded_users)
    num_lost = num_users - num_decoded
    plr = num_lost / num_users
    throughput = num_decoded / num_slots

    return {"plr": plr, "throughput": throughput}


def monte_carlo(num_users, num_slots, degree_dist, trials=500, seed=0):
    rng = np.random.default_rng(seed)
    plrs, throughputs = [], []
    for _ in range(trials):
        s = int(rng.integers(0, 2**31 - 1))
        res = simulate_frame(num_users, num_slots, degree_dist, seed=s)
        plrs.append(res["plr"])
        throughputs.append(res["throughput"])
    return {
        "G": num_users / num_slots,
        "plr": float(np.mean(plrs)),
        "throughput": float(np.mean(throughputs)),
    }


num_slots = 200
degree_dist = {2: 0.5, 3: 0.28, 8: 0.22}
trials = 800

loads = np.arange(0.1, 1.01, 0.05)
results = []
for g in loads:
    num_users = max(1, int(round(g * num_slots)))
    m = monte_carlo(num_users, num_slots, degree_dist, trials=trials)
    results.append(m)

Gs = [r["G"] for r in results]
plrs = [r["plr"] for r in results]
thrs = [r["throughput"] for r in results]

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

ax = axes[0]
ax.semilogy(Gs, [max(p, 1e-5) for p in plrs], marker="o", color="crimson")
ax.set_xlabel("Channel load G = users / slots")
ax.set_ylabel("Packet Loss Rate (PLR)")
ax.set_title(f"IRSA-SIC: PLR vs Load\n(num_slots = {num_slots})")
```

```
ax.grid(True, which="both", alpha=0.3)

ax = axes[1]
ax.plot(Gs, thrs, marker="s", color="steelblue", label="Achieved throughput")
ax.plot(Gs, Gs, linestyle="--", color="gray", label="Ideal T = G")
peak_idx = int(np.argmax(thrs))
ax.scatter([Gs[peak_idx]], [thrs[peak_idx]], color="darkorange", zorder=5,
           label=f"Peak T = {thrs[peak_idx]:.4f} @ G = {Gs[peak_idx]:.2f}")
ax.set_xlabel("Channel load G = users / slots")
ax.set_ylabel("Throughput T (decoded packets / slot)")
ax.set_title(f"IRSA-SIC: Throughput vs Load\n(num_slots = {num_slots})")
ax.legend()
ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig("irsa_sic_result.png", dpi=150)
plt.show()
```

Listing C.1 – Standalone IRSA/SIC simulation (irsa_sic_main.py)

```
# Baselines evaluated with the same functions, frame size, loads, trials and seed
import numpy as np
from irsa_sic import monte_carlo, build_transmit_matrix  # the four functions of Listing C.1

num_slots, trials = 200, 800
loads = np.arange(0.1, 1.01, 0.05)
schemes = {"FSA": {1: 1.0}, "CRDSA": {2: 1.0}}

for name, dd in schemes.items():
    for g in loads:
        n = max(1, int(round(g * num_slots)))
        m = monte_carlo(n, num_slots, dd, trials=trials)
        print(name, f"{m['G']:.2f}", f"{m['throughput']:.4f}", f"{m['plr']:.3e}")

# IRSA without SIC: a user is decoded only if it has a replica in an initial singleton slot
dd = {2: 0.5, 3: 0.28, 8: 0.22}
for g in loads:
    n = max(1, int(round(g * num_slots)))
    rng = np.random.default_rng(0)
    thr, plr = [], []
    for _ in range(trials):
        s = int(rng.integers(0, 2**31 - 1))
        M = build_transmit_matrix(n, num_slots, dd, np.random.default_rng(s))
        single = np.where(M.sum(axis=0) == 1)[0]
        dec = np.unique(np.where(M[:, single] == 1)[0])
        thr.append(len(dec) / num_slots)
        plr.append(1 - len(dec) / n)
    print("IRSA_noSIC", f"{n/num_slots:.2f}", f"{np.mean(thr):.4f}", f"{np.mean(plr):.3e}")
```

Listing C.2 – Baseline schemes under identical conditions (baselines.py)